SyteLine: Fundamentals of Advanced Planning and Scheduling (APS) Training Workbook

SyteLine
April 20, 2024
Course Code: 01_0012105_IEN3734_SYT

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Contents

About this workbook

Welcome to this Infor Education course! We hope you will find this learning experience enjoyable and instructive. This Training Workbook is designed to support the following forms of learning:

- Classroom instructor-led training with an Infor certified instructor
- Virtual classroom instructor-led training with an Infor certified instructor
- Self-directed learning through Infor Campus

This Training Workbook is not intended for use as a product user guide.

Workbook design

This Training Workbook contains both conceptual information to introduce topics and step-by-step procedural instructions for practical application of those concepts.

Symbols and notes are provided throughout this Training Workbook for ease of reference. Refer to the *Symbols used in this workbook* section below to familiarize yourself with these symbols.

The PDF bookmarks pane provides quick and easy navigation between lessons, topics, and appendices, when needed.

Instructor-led training (ILT)

If you are taking this course as ILT, your instructor will provide details on accessing the Infor Education Training Environment needed to complete the student exercises. Your instructor will also assign you and other students an account login and password from Appendix A of this Training Workbook.

Some exercises in this workbook cannot be performed in the live Training Environment that is shared by multiple students during class. In these instances, a hyperlink to a recorded simulation is provided in the Training Workbook for your reference. Your instructor may also perform the exercises one time in the system on behalf of the class, if needed. Do not attempt to complete the steps for any of these demos in the system. Doing so could adversely affect the Training Environment, the intended flow of the course, and the success and quality of the course for all students in the class.

Self-directed learning (SDL)

Some SDL courses offer Lab on Demand in addition to simulations. If you are taking this course through Infor Campus as SDL, and the course offers Lab on Demand, refer to the Lab on Demand screen in the self-directed learning course for course environment information. The Lab on Demand screen includes instructions and logins to launch and access the corresponding Infor Education Training Environment as well as logins and passwords required for completion of course exercises and demos.

The exercises and demos in this course build upon each other as they prepare the system for subsequent exercises and demos. If you are taking this course as SDL with Lab on Demand, you must complete all of the exercises and demos in the order they are presented in the Training Workbook. This ensures you will achieve the expected results and a successful course outcome.

Instructor-recorded presentations and simulations (SDL only)

If you are taking this course as SDL, there may be instructor-recorded presentations and/or simulations available to assist you.

If instructor-recorded presentations are available, a hyperlink to the recording will be included on the first page of each corresponding lesson of the interactive workbook on the Training Workbook tab of the self-directed learning course.

If simulations are available, the demos and exercises throughout the interactive Training Workbook will include hyperlinks to simulations that allow you to view and/or practice the execution of the demo or exercise. These same simulations are also accessible via the Demonstrations tab of the self-directed learning course. Please note that the data used in some simulations may not exactly match the data provided in the Training Workbook.

Learning Libraries

Learning Libraries in Infor Campus include learning materials that are available to you online, anytime, anywhere. These materials can supplement instructor-led training, providing you with additional learning resources to support your day-to-day business tasks and activities.

Please note that if you accessed this Training Workbook directly via a Learning Library, you will not have access to the Infor Education Training Environment that is provided with all instructor-led and most self-directed learning course versions, as referenced above. Therefore, you will not be able to practice the exercises in the specific Training Environment for which the exercises in this Training Workbook were written.

Symbols used in this workbook

Exercise	Your notes	Question
Demo	Important note	Answer
Scenario or Discussion	Critical note	Task simulation
For your reference	Simulated activity	

Course overview

Estimated time

0.5 hours

Learning objectives

Upon completion of this course, you should be able to:

- Explain the basics of how Advanced Planning and Scheduling (APS) works.
- Identify important APS parameters and setup requirements.
- Explain how to model the site to use APS.
- Explain how to plan the site.
- Define next steps for implementing APS.

Topics

- Course description and agenda

Course description and agenda

This course teaches you the essential requirements for configuring Advanced Planning & Scheduling (APS) and setting up master data for APS. You will learn the critical forms, fields, procedures, and processes used to run and administer APS.

This training is for version 10.20 and all previous versions.

Course duration

4 hours

Prerequisite courses

- CloudSuite Industrial: Navigating the User Interface
- CloudSuite Industrial: Foundation
- SyteLine: Advanced Planning and Scheduling (APS) Overview and Concepts

Prerequisite knowledge

To optimize your learning experience, Infor recommends that you have the following knowledge prior to taking this course, based upon your role:

- CloudSuite Industrial: Using Customer Service/Order Entry
- CloudSuite Industrial: Using Bills of Materials & Engineering Change Notices
- CloudSuite Industrial: Using Job Orders

Audience

- Customer user
- Pre-sales consultant
- Business consultant
- Technical consultant
- Support
- System administrator

System requirements

- Infor SyteLine Training Environment

Reference materials

SyteLine reference materials are available from the following locations:

- SyteLine Help menu
- Infor Concierge®

Course agenda

The agenda below details the contents of this course, including lesson-level learning objectives and supporting objectives.

Lesson	Lesson title	Learning objectives	Estimated time (hours)
Course overview		Review course expectations.	0.25
1	How APS works	Explain the basics of how Advanced Planning & Scheduling (APS) works. • List the key APS principles. • Describe the three-pass planning method. • Identify the functional areas involved with APS.	1.0
2	APS parameters and setup	Identify important APS parameters and setup requirements. • Describe key parameters used in planning and scheduling. • Identify the different planning modes available with APS.	0.5
3	Modeling the site for planning	Explain to how to model the site to use APS. • Describe how resources are used in APS. • Identify the different types of records used in APS. • Explain the importance of an item's bill of materials (BOM).	1.0
4	Run APS and view the plan	Explain how to plan the site. • Describe the launch control process. • Explain how to generate the Material Planner Workbench. • Explain how to plan the site. • Identify APS outputs used for analysis. • Describe the basics of troubleshooting the plan.	0.5
5	Next steps	Define the next steps for implementing APS. • Identify implementation preparation steps. • Describe critical success factors. • List additional APS learning opportunities.	0.5
Course summary		Debrief course.	0.25

Appendices

This section contains information that is not part of the instructional content of this course, but it provides additional related reference information.

Appendix	Appendix title	Content description
Appendix A	User accounts	This appendix provides a reference for student and instructor login credentials and if applicable, assigned exercise data.
Appendix B	Check your understanding answers	This appendix provides answers to the check your understanding questions found at the end of each lesson.
Appendix C	Setting up APS	This appendix provides the recommended settings for the Planning Parameters form along with additional setup information.
Appendix D	Running the APS planning activity	This appendix provides reference information about running the APS planning task.

Lesson 1: How APS works

Estimated time

1 hour

Learning objectives

After completing this lesson, you will be able to explain the basics of how Advanced Planning and Scheduling (APS) works. In this lesson, you will:

- List the key APS principles.
- Describe the three-pass planning method.
- Identify the functional areas involved with APS.

Topics

- Overview
- Key principles of APS
- Three planning passes
- Planning dates
- The process of planning with APS
- Functional areas involved with APS

Overview

Planning is the process of harmonizing supply with external demand. External demand for any given site can include:

- Customer orders
- Forecasts of future customer orders
- Project resources
- Safety stock
- Transfer orders from other sites

You meet those demands with quantities that are not yet allocated from:

- On-hand inventory
- Expected deliveries (active work orders, purchase orders, transfer orders, and planned orders)
- New planned orders (to make, buy, or transfer items)

APS generates real-time projections of when you can complete orders by comparing all demands against a long-term plan. The system views the current status of inventory levels, forecasts, job schedules, purchase order (PO) due dates, customer orders, etc., and creates planned orders accordingly to satisfy the demands. You then "firm" the planned orders into POs, purchase requisitions, job orders, production schedules, or transfer orders.

The way APS matches supply with demand is to:

- Plan each order separately
- Use up to three general planning passes per order
- Create new planned orders only when there are no supplies

APS will create new planned orders down through the lowest level of the BOM if needed.

How a single demand moves through planning and production

The life cycle for a single demand usually proceeds through the planning and scheduling functions as shown in the diagram below.

This process typically includes the following steps:

- Order entry user promises the order to customer
  - Demand can be incrementally planned.
- Planner runs APS Planning activity
  - APS generates planned orders (PLNs) as needed to fill unsatisfied demand quantity.
  - Planner firms the PLN into a job and releases it.
- Scheduling activity schedules job and generates dispatch list
- Job is executed on shop floor according to dispatch list
- Order is shipped to customer

APS daily activities

The daily tasks related to planning and scheduling include the following (some of these tasks occur in parallel).

- Run the APS Planning activity
  - The task can be placed on the Background Queue to run nightly, when few users will be making changes to records that affect the plan or schedule.
- Create demands/promise orders during order entry and other daily activities
  - Use incremental planning functions to project completion dates and insert demand into current plan.
- Analyze planning output and determine causes for any lateness
- Respond to exception messages
  - Exceptions Report
  - Planning Detail form

- Run Material Planner Workbench to determine when to generate/release job order or purchase order to ensure that an item is available when needed
- Firm PLNs from Material Planner Workbench form or Planning Detail form
- Execute production according to the dispatch list
- Record status of operations for costing and scheduling purposes
    - Setup time, move time, etc.

Key principles of APS

As you learned in the APS Overview and Concepts course, if you want to use APS to help you make and keep promises using a lean philosophy, you will adjust your practices, policies, and procedures to support that. This means you need to:

- Have business processes in place to ensure you provide valid data
- Model your planning environment accurately, both materials and capacity
- Plan the work and then work the plan

The four critical APS principles and their accompanying practices are detailed in the table below.

Principles	**Practices**
Keep your data accurate	• Record labor and material transactions timely and accurately • Keep inventory records accurate • Close job orders in a timely manner • Keep routings and BOMs up to date • Give purchased items accurate lead times • Depict work center capacity realistically
Follow the plan	• Follow the work center dispatch lists • Release orders on time and according to APS release dates (launch control) • Release job orders only when you have all the materials for them in inventory (launch control) • Don't second guess the system by putting up shop orders to hedge capacity or projected needs
Get ahead of the game	• Review bottleneck, overcapacity, and projected late order charts daily to anticipate problems • Limit the number of drop-ins • Make realistic promises when taking customer orders
Keep it simple	• Limit the number of constrained resources • Simplify what you model in the system: Use the 80/20 rule

Three planning passes

When APS plans a demand, it first searches backward from the due date to allocate the needed quantity (this is “pull planning”), and calculates the start date for the demand.

In some cases, the system may not be able to plan the demand in the time between the due date and the current date, and the start date is calculated to be in the past. In this situation, the system instead plans the demand forward from the current time (this is “push planning”).

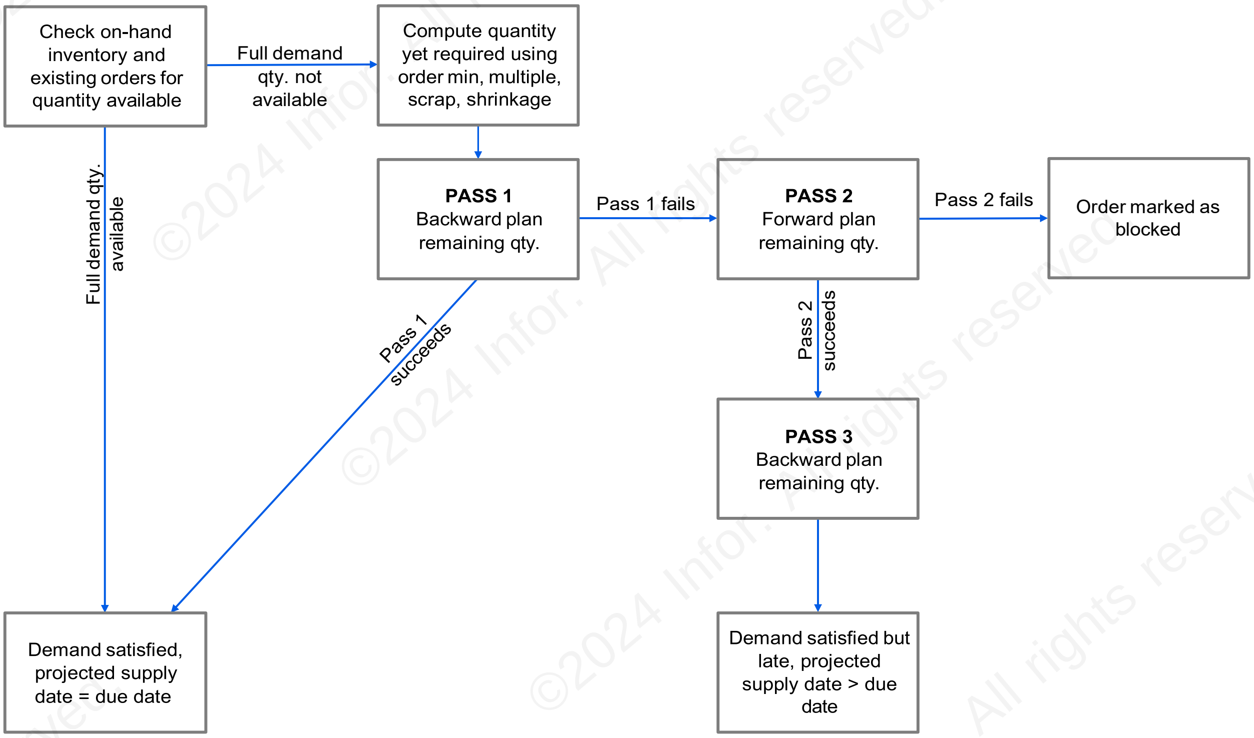
Then, after APS plans a push pass, it plans an additional pull pass, this time using the date calculated during the push pass.

This is known as the three-pass method, described below.

Pass	Method	Description
1 – Pull pass	Backward plan and schedule	APS goes through the questions in the order shown below: • Do we have stock available? • If not, do we have quantities available on expected deliveries? • If not, can we make or purchase it in time?
2 – Push pass	Forward plan and schedule	APS only goes to pass 2 if pass 1 fails. In this pass, APS starts with the current date and time and schedules forward, calculating a worst-case projected date. We don’t see this date because APS immediately does pass 3 after pass 2.
3 – Pull pass	Backward plan and schedule	APS only does this pass if it did pass 2. In this pass APS backward plans and schedules to: • Remove any slack in the plan and schedule. • Try to improve the projected date over the worst case calculated in pass 2. The system can sometimes improve the projected date because the forward and backward planning passes use different algorithms.

Planning process diagram

In order to set up APS and then use it to troubleshoot your projected late shipments, you must understand how APS plans. The diagram below summarizes the method APS uses to backward plan and schedule supply to meet demand. As you can see, APS may take up to three planning passes to accomplish this.



Below you will find a more detailed explanation of the visual above.

1. APS checks available supply
   a. APS looks to see if there are any quantities on-hand or existing orders that are available to meet the order requirements.
   b. If there are quantities available, APS allocates those quantities for that specific order and returns a projected date equal to the order's due date. At this point, APS has finished planning this order.
2. APS backwards plans from the order's due date.
   a. If there are not enough available on-hand or on-order quantities, APS will see if you have the time, materials (or the ability to get the materials), and resources to make or purchase the remaining quantity by the due date. For purchased and transferred items, APS looks at the lead times found on the Items form to see how long it will take to get the materials in. For manufactured items, APS uses the item's BOM to calculate the time, resources, and materials needed.
   b. If you can make or purchase what you need to fill the order by the due date, APS creates all the necessary PLNs with their required release dates and returns a projected date equal to the order's due date. At this point, APS has finished planning this order.
   c. If you cannot ship by the due date, that is, if you need more time than you have to purchase or make the items on the order, APS goes to the second planning attempt.
3. APS forward plans the order requirements from the current date and time.
   a. APS looks to make sure you have available components. If you do, it will allocate them. If not, it will create PLNs to make or buy the materials. Again, for purchased and transferred items, APS looks at the lead times found on the Items form to see how long it will take to get the materials in. For manufactured items, APS uses the item's BOM to calculate the time, resources, and materials needed.

b. APS forward schedules your resources from the point when APS calculates all the components will be available for the operation.
c. The result of this second planning pass is a projected order finish date that is feasible. However, because the algorithms APS uses for its forward planning are more conservative than those it uses for its backwards planning, this is a "worst case" date. Before APS finishes, it will do one more backwards plan to see if it can improve the projected date and take out any slack according to just-in-time (JIT) principles.
d. If APS can’t plan the order in the planning horizon, the order is marked as “blocked” and stops planning the order. No projected dates, resource or material allocations, or PLNs will be created for the order.

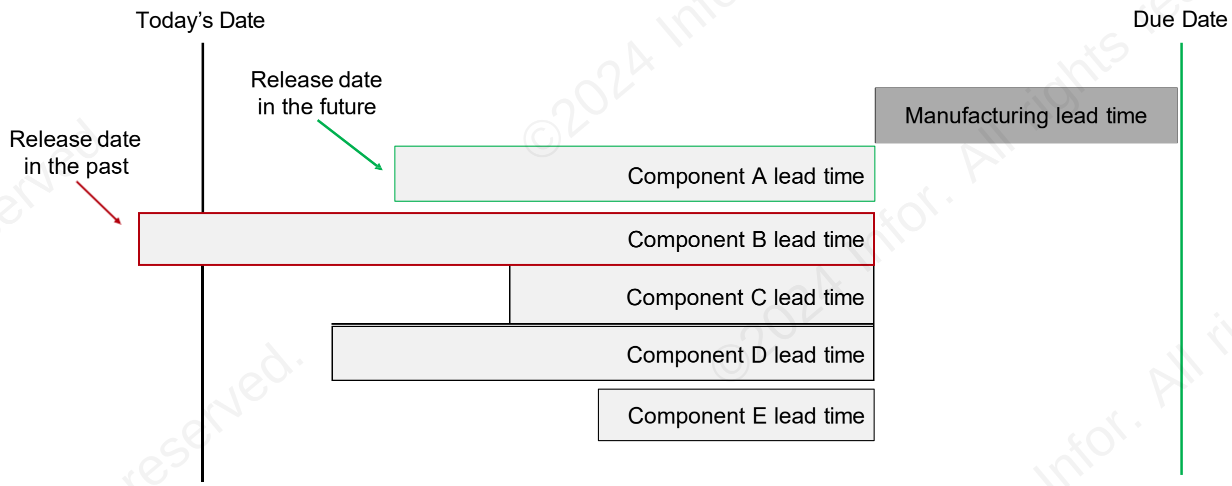
4. APS backwards plans from the projected date calculated in the second pass and tries to improve it.
   a. Using the backwards planning algorithms, APS takes out any slack in the PLNs it has created to fill this demand, making sure you do not order too early and end up carrying unnecessary inventories or unnecessarily tie up resources.
   b. APS also tries to improve the projected date.
   c. At this point, APS has finished planning this order and returns the best calculated projected finish date.

Planning dates

APS uses an order's due date as its target. But the supply for an order is tied to the projected date of the supply. Your goal is to make sure the projected date of the supply matches the due date of the order. This is a critical concept to understand because this planning behavior of separating the actual supply date from the due date has significant ramifications. And if you've been using Material Requirements Planning (MRP), it will require a new way of thinking because MRP doesn't do this.

Planning with due date only

MRP backward plans from the due date. It never assumes a supply will come after the due date. For example, look at the diagram below. It shows MRP planning to the due date. Notice that you have enough lead time for everything except Component B. However, MRP doesn't change any of the other supply plans because of that. It simply assumes you will expedite Component B, so it arrives by the time the job needs to start.



If you are coming from an MRP background, you are probably accustomed to using past-due release dates as the warning that you are at risk of shipping late. You have also probably used them to tell you which supplies were the most critical to expedite—the farther in the past the release date, the higher priority.

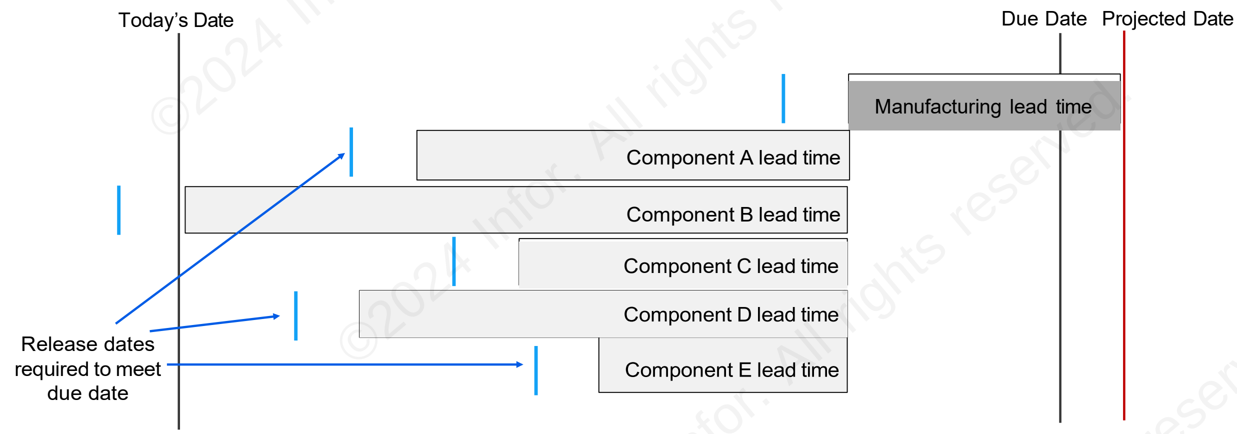
Because MRP plans into the past, planners sometimes adjust by inflating lead times to make sure the components are in stock by the time the jobs need to start. This means that component supply may arrive earlier than needed and sit around longer than necessary. And if the lead times are significantly inflated, it can also lead to unnecessary expediting.

It's important to realize that inflating lead times only causes problems when planning with APS.

Planning with a due date and projected supply date

APS targets an order's due date. But it also recognizes that there might be constraints that push supply dates past a due date. For example, APS doesn't consider releasing orders in the past as feasible, so it won't generate plans based on this. The minute APS sees that you don't have enough lead time to get a component, it will push out the projected date for the whole order.

The diagram below illustrates how APS would handle the same situation that we looked at above. The first pass of APS identifies a release date for Component B in the past, which is not possible. Therefore, it pushes the projected date of the whole order out to accommodate the Component B lead time.



Notice that APS then backward plans everything from that projected date. When APS pushes the projected date out to accommodate Component B, it adjusts the supply plans for all the other components too. APS drags all those supplies out because it's planning in a JIT manner. There's no reason to release other components if they can't be put to use until Component B arrives.

If your lead times are inflated, APS will push your projected supply dates out more frequently than is necessary. And it will drag all the other supplies out as well. To avoid the issues this creates, you will need to make sure your lead times for both purchased and manufactured items are accurate, as specified in the first key APS principle.

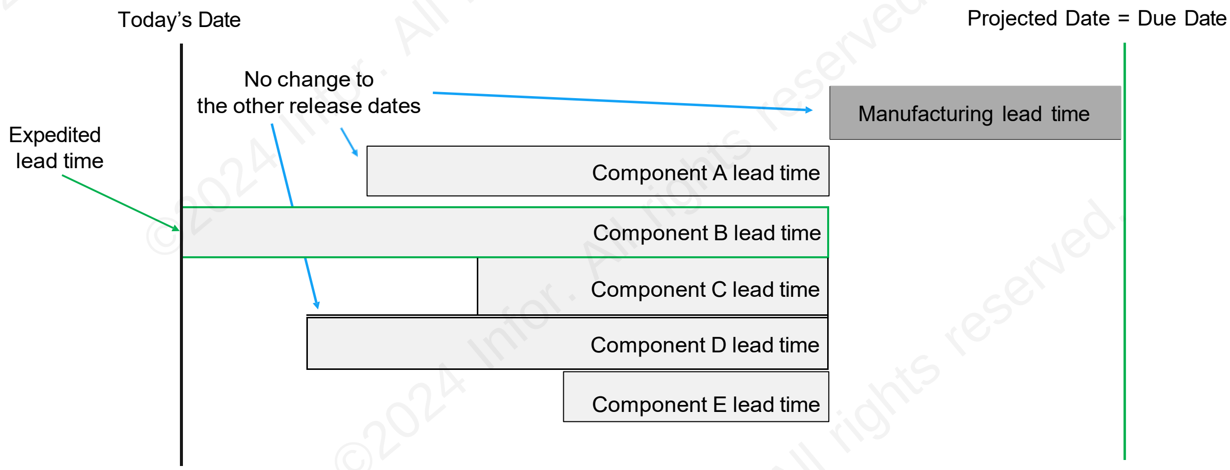
Changing your constraints to reflect your flexibility

What if your lead times are reasonably accurate, and you'd rather try expediting Component B first before pushing out the projected date and all related supplies? After all, if you can usually expedite Component B, you don't want to miss the release window for your other components.

As we reviewed in the APS Overview and Concepts course, if you can flex your lead times and capacity and would rather do that instead of pushing a projected date out, then you need to let APS know.

For example, if Component B is a purchased part, you can tell APS that it's possible to expedite it. If you do, when APS sees that you don't have enough time with the normal lead time, it will use the expedited lead time and not change any of the other supply plans.

Look at the following diagram. APS assumes you can expedite Component B and leaves all the other supply plans as they were.



In this situation, APS will also give you an exception message, telling you to expedite the supply for Component B. So, you can accommodate expediting with APS, but you're going to do it in a different way than you would with MRP.

You can also accommodate flexing your capacity. However, even if you can flex lead times and capacity and set APS up to do so, APS will still always plan with a separate projected supply date. And if there is ever a time when you can't flex your lead times or capacity enough to meet the due date, APS will push the projected supply date out and drag all the other supply plans out with it.

Why will it do this?

Because it's trying to give you a feasible plan. If you can't flex enough to meet the due date using those standard resolutions, you can't. And it's better to know that as soon as possible instead of waiting until your order is late.

Different warning signals

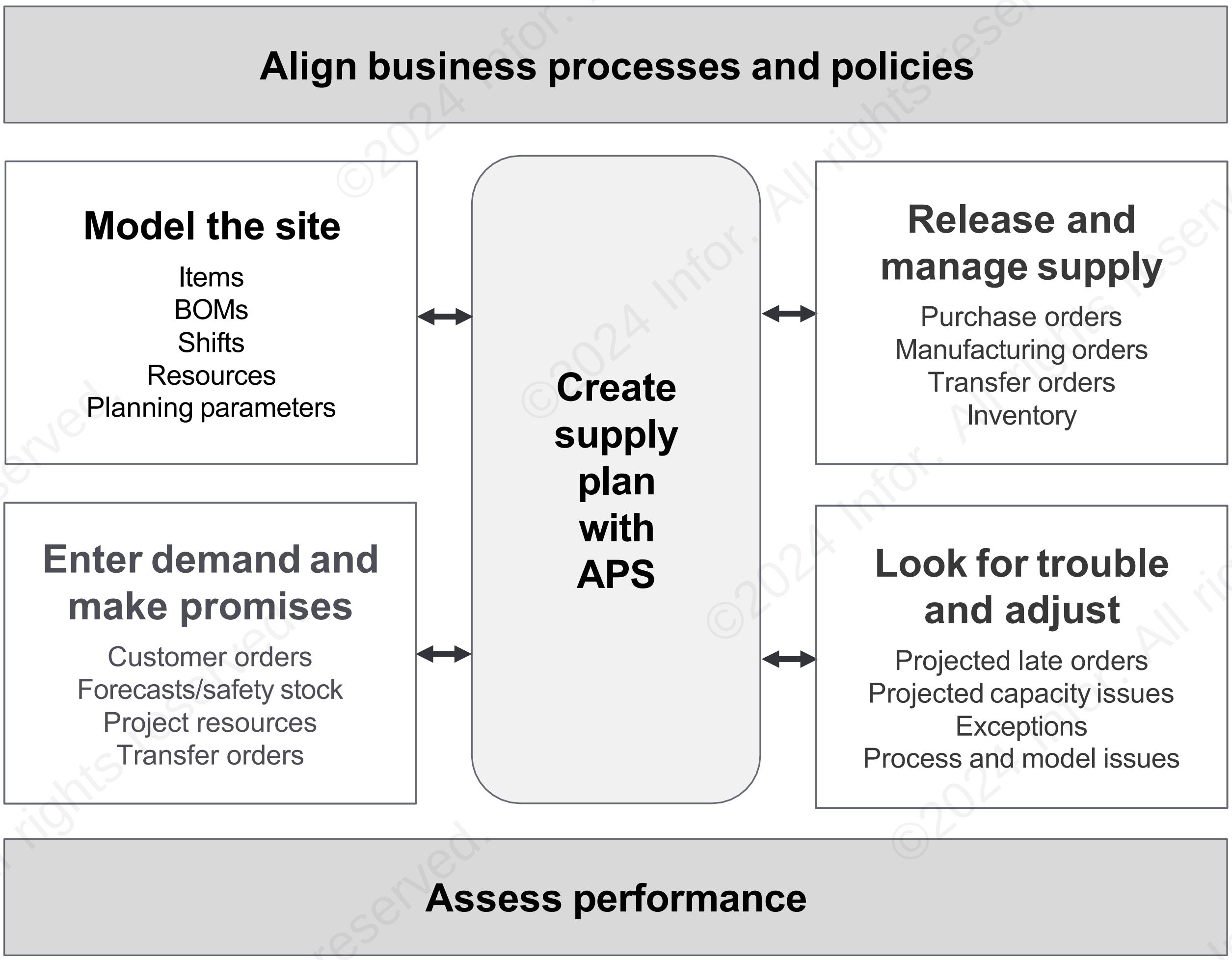
Note that with APS there is no planning in the past. If you're coming from an MRP environment, that means you won't be able to use release dates in the past as a warning signal. They don't exist with APS because APS won't plan release dates in the past.

Instead, if APS is set up to assume you'll flex your purchased part lead times and capacity, it will provide you with information about when you need to take action to expedite purchase orders or add resources. You can also use APS output to spot supply orders that are projected to finish late and evaluate what's causing the delay. This tells you when to take special measures to resolve the issue or work out a more feasible due date with the customer.

The process of planning with APS

APS has a lot of moving parts. If you immediately dive into the details, you will get lost. However, if you first take some time to familiarize yourself with the big picture of how APS works, you'll find it much easier to understand the rest of the class.

The diagram below illustrates the APS framework. There are seven elements to this framework. Your organization needs to develop skills to effectively address all seven of these elements.



Most of these elements require that your organization take action. Most also generate application activity. The table below summarizes these actions and activities.

Element	Activity	What you do	What APS does
1	Align the elements of your company to support planning success.	Verify your goals and the goals of APS are in harmony. Identify and adopt the processes, policies, and procedures that ensure accurate inputs and help achieve your goals.	

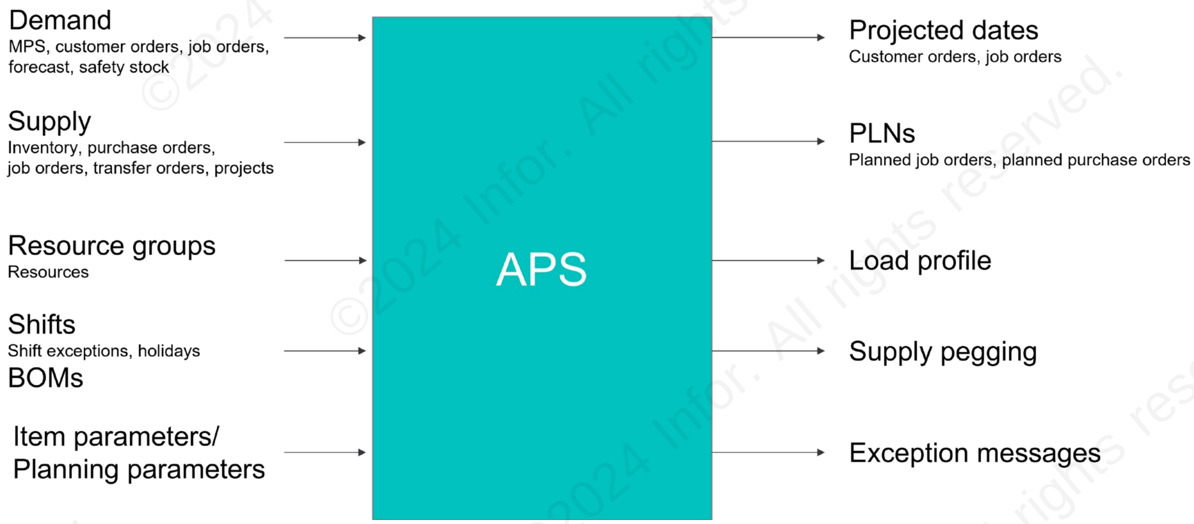
Element	Activity	What you do	What APS does
2	Model the site for planning with APS.	Provide accurate input, keeping it as simple as possible.	
3	Enter demand and make promises to customers.	Make realistic promises to customers and create reasonable forecasts.	Provide item availability information. Plan new demands in real-time.
4	Create a supply plan to meet demand.		Create the production plan and the procurement plan in a JIT fashion.
5	Release supply orders according to the APS orchestration.	Review the list of new purchase orders, transfer orders, and job orders APS is suggesting you release, take any special actions required, then release them in a JIT fashion.	Provide a list of planned purchase order and job order supplies you need to release today. Indicate which of the planned purchase order supplies APS is expecting you to expedite. Indicate which planned jobs are ready to be released and which are missing components.
6	Look for trouble and adjust to ensure supply stays on track to meet your due dates.	Review APS alerts and take appropriate action to resolve any issues.	Provide alerts to tell you when: • You are projected to ship customer orders late. • You are projected to finish manufacturing orders late. • APS is expecting you to flex your capacity. • APS is expecting you to move in existing supply orders. • You have supply orders that are not needed.
7	Assess performance.	Track key performance indicators for each of the other parts of the framework.	

APS inputs and outputs

As you can see in the framework above, your job is to:

- Make sure APS has accurate inputs.
- Review and act on the APS outputs.

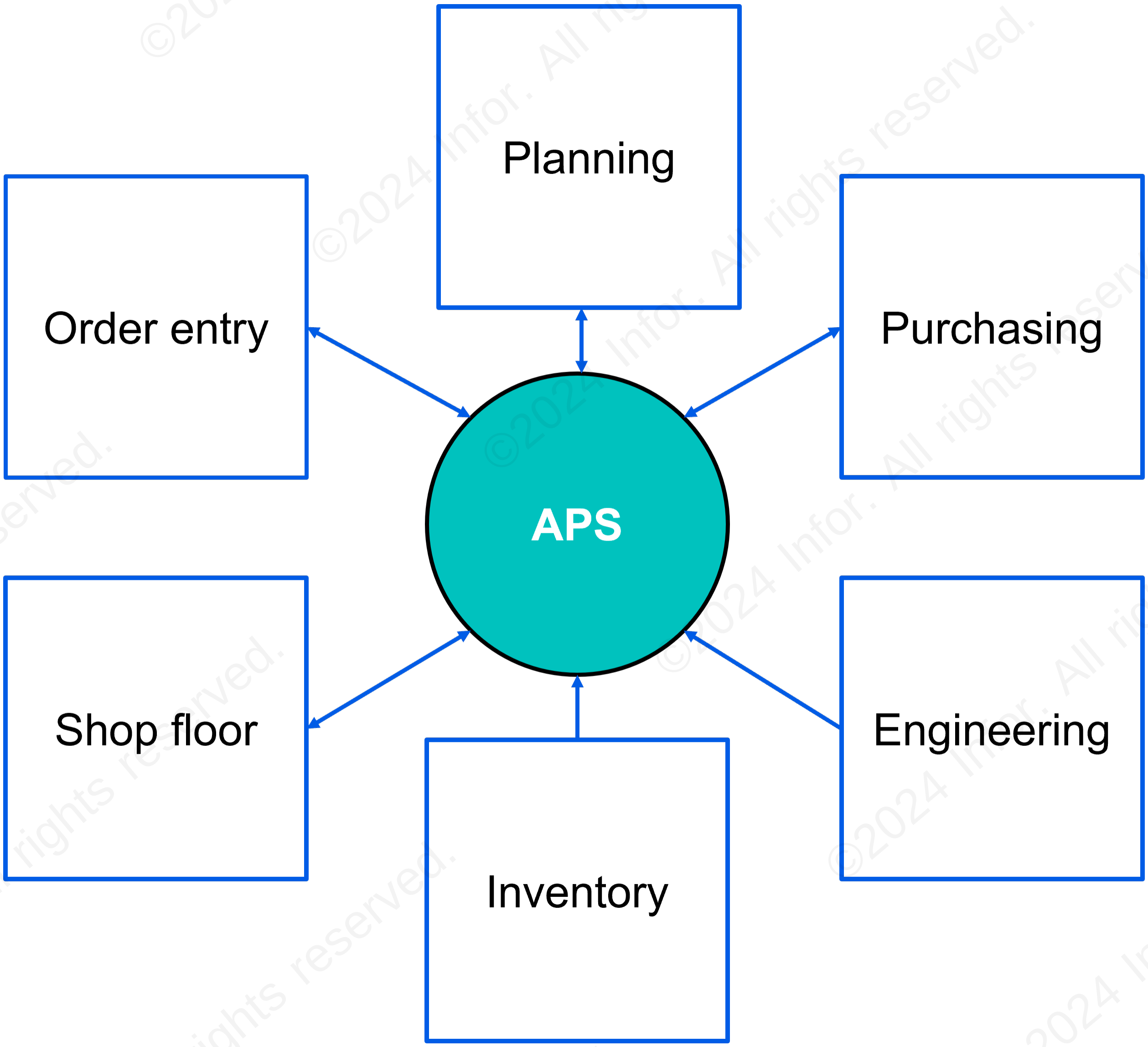
The following diagram identifies the primary inputs and outputs.



Notice that a number of different areas of the company will need to be involved in the process to make it work. It's important to clarify what parts are involved and what each one does.

Functional areas involved with APS

There are six main areas of the company that are critical for the planning process, shown in the diagram below. Each of these areas must make sure they are providing valid inputs to APS. Four of these areas also work with the outputs APS produces.



Let's look at what each does on a daily basis in a little more detail.

Daily tasks

The six main areas that work with APS and what each does on a daily basis are described in the tables below. As you can see, all of these areas of the company need to be aligned for the planning process to work.

Order entry

Daily tasks for the order entry/customer service team include the following:

Task	Description
Enter demand	Enter customer orders
Respond to APS suggestions	Make good promises
Update the plan	Plan individual orders

Planning

The planning group is responsible for a variety of daily tasks for APS as outlined below.

Task	Description
Enter demand	• Develop and enter forecasts with input from the shop floor and sales • Develop and enter Master Production Schedule (MPS) orders if applicable
Respond to APS suggestions	• Create new job orders suggested by APS • Review and react to exception messages
Look for trouble and resolve issues	• Perform late order analysis • Identify and resolve capacity and material shortages
Keep data accurate	• Maintain item and planning parameters • Work with purchasing, shop floor, engineering, and inventory to keep item, resource, and BOM data accurate
Update the plan	• Plan the site • Review and resolve planning errors, warnings, and blocks
Report performance metrics	• Track key performance indicators for on-time shipment, throughput, expediting, etc.

Purchasing

Individuals in a purchasing role will handle the tasks shown below.

Task	Description
Respond to APS suggestions	• Create new POs suggested by APS • Review and react to exception messages
Look for trouble and resolve issues	• Identify material shortages • Expedite purchased parts
Keep data accurate	• Update PO availability and purchased part lead times

Shop Floor

The daily tasks for the shop floor are shown below.

Task	Description
Respond to APS suggestions	• Make final adjustments to the APS schedule • Communicate daily dispatch to floor • Work the schedule
Look for trouble and resolve issues	• Identify job orders projecting late • Resolve late job order issues
Keep data accurate	• Post time and materials • Close jobs

Engineering

The engineering function is primarily responsible for data accuracy and validity.

Task	Description
Keep data accurate	• Create and update BOMs • Work with planning and the shop floor to ensure BOMs, resources, and resource groups are accurate

Inventory

Like engineering, the inventory team must focus on data integrity.

Task	Description
Keep data accurate	• Post material issues, shipments, receipts, and moves • Count and update inventory records

Key forms used in the tasks

Below we list the key forms that are used in the various APS activities and the areas of the company that typically use them. There are other forms that are used in the planning process, but these are the key forms, and the ones we will focus on in this course.

It is not critical to review all of these forms in detail now. However, you may find it helpful to refer back to this list as you work through the course and start using APS in your daily tasks.

Modeling the environment

These forms are used for setting up your planning model.

Form	OE	PL	PR	SF	EN	Notes
Planning Parameters		X				Establish critical global settings for planning.
Scheduling Shifts		X		X		Define scheduling shifts, including PCAL, as well as downtime and overtime.
Shift Exceptions by Group		X		X		Define downtime and overtime for multiple resources.
Holidays		X		X		Define downtime for all resources.
Resources		X			X	Define labor and machine resources.
Resource Groups		X		X	X	Define sets of resources that can do the same thing.
Work Centers		X			X	Define the default resource group to be used on BOM. APS ignores work centers, but setting the default can make creating BOMs easier.
Items		X	X	X	X	Define many critical planning parameters.
Current Operations		X		X	X	Define critical operations and sequence; normally specify only one resource group.
Current Materials		X		X	X	Define the materials required by an operation.

Key: OE=order entry, PL=planning, PR=purchasing, SF=shop floor, EN=engineering

Planning the whole site

Use these forms for setting up, running, and monitoring the planning activity.

Form	OE	PL	PR	SF	EN	Notes
Infinite APS or APS Mode		X				Define whether you are going to allow APS to schedule finitely or infinitely.
APS Planning		X				Run the planning process for the whole site, which is usually performed as a background task once a day.
Background Task History		X				Review status of the planning activity.
APS Planning and Scheduling Messages		X				Identify errors in the planning run.
Planning Detail	X	X	X	X	X	Monitor time phasing of demand and supply.

Key: OE=order entry, PL=planning, PR=purchasing, SF=shop floor, EN=engineering

Planning an individual order

To incorporate an individual order into the plan, use these forms.

Form	OE	PL	PR	SF	EN	Notes
Customer Orders and Lines	X	X				Define demand requirements. Verify in real time your ability to meet demand if customers require that information while talking to you.
Forecasts		X				Define expected demand requirements.
Master Production Schedule		X				Define demand requirements (only used when you don't want APS to plan to customer orders and forecasts).

Key: OE=order entry, PL=planning, PR=purchasing, SF=shop floor, EN=engineering

Releasing planned supplies

These forms are used for releasing supply components in accordance with the APS plan.

Form	OE	PL	PR	SF	EN	Notes
Material Planner Workbench Generation		X	X			Select new order suggestions to re-view, which enables launch control of orders.
Material Planner Workbench		X	X			Review and convert APS order sugges-tions into actual orders, which enables launch control of orders.
Purchase Orders and Lines			X			Define detailed supply information.
Job Orders		X		X		Define detailed supply information.
Supply Usage APS		X	X	X		View tie between supply and demand.

Key: OE=order entry, PL=planning, PR=purchasing, SF=shop floor, EN=engineering

Troubleshooting

There are many forms which can help you identify and analyze issues with the plan. These are some of the most commonly used forms.

Form	OE	PL	PR	SF	EN	Notes
Demand Summary APS		X		X		Identify customer orders and jobs which are projected to be late.
Demand Detail APS		X		X		Identify plan for each order and what's causing lateness.

Form	OE	PL	PR	SF	EN	Notes
Resource Group Utilization APS		X		X	X	Identify over capacity resources.
Resource Group Load Profile APS		X		X	X	Identify over capacity resources and loads by type.
Resource Group Plan		X		X	X	Identify detailed information about loads in a given interval.
Planning Detail		X	X	X		Identify all exception messages and view shortages over time.
Component Shortages APS		X		X		Identify jobs that do not have all the necessary components.
Inventory Summary		X	X	X		Identify items causing delays.
Alternative Summary APS		X	X	X		View dashboard of the key red flags—orders projecting late, over ca- pacity resources, item shortages.

Key: OE=order entry, PL=planning, PR=purchasing, SF=shop floor, EN=engineering

Reviewing and printing the shop floor schedule

Use these forms to review, print, and adjust the shop floor schedule.

Form	OE	PL	PR	SF	EN	Notes
Dispatch List Report				X		Print daily dispatch for resources.
Resource Sequencing and Resource Group Sequencing				X		Adjust the sequence of work to be completed by a resource or resource group. **Note:** Complete this before you print the dispatch.

Key: OE=order entry, PL=planning, PR=purchasing, SF=shop floor, EN=engineering

Check your understanding

?

Match each of the following essential APS practices with the key principle the practice supports.

!

Essential APS Practice	Key APS Principle
Make realistic promises when taking customer orders	Keep data accurate
Keep routings and BOMs up to date	Follow the plan
Limit the number of constrained resources	Get ahead
Release job orders only when you have all needed materials in inventory	Keep it simple
Use the 80/20 rule to decide what to model	Follow the plan
Record transactions timely and accurately	Keep it simple
Review bottlenecks, capacity issues, and projected late orders daily	Keep data accurate
Release orders on time and according to APS release dates	Get ahead

?

When does APS perform forward planning?

!

(a) On every demand
(b) When the demand due date is within a specified number of days from today's date
(c) When the demand due date cannot be met because the supply dates are later than the due date
(d) After every pull pass

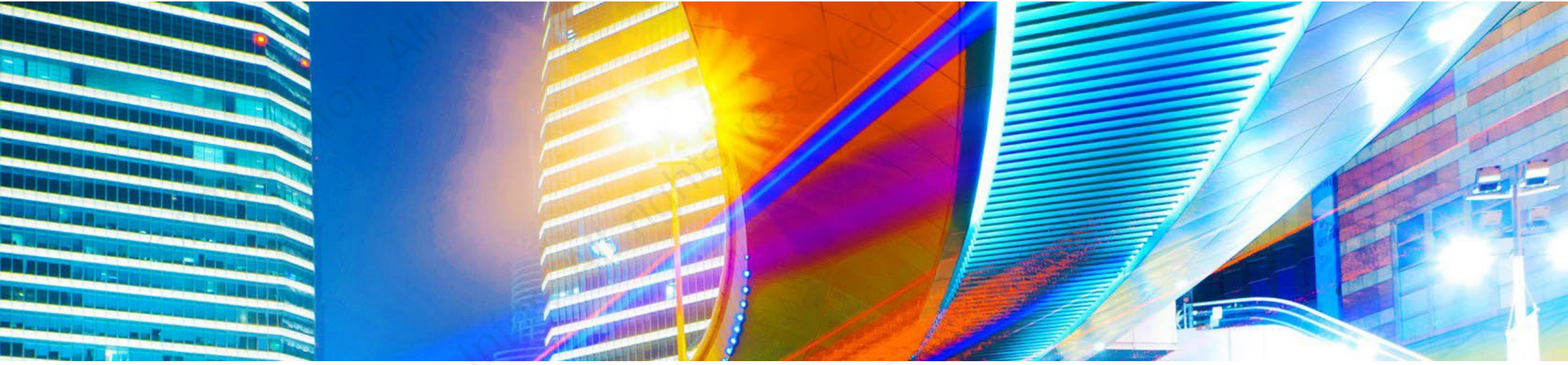
To meet the due date on a customer order, APS calculates you will need a part in three days, but there's no inventory available, and it will take 10 days to get the part in. APS will plan the customer order's projected date:

(a) Before the order due date
(b) After the order due date
(c) Equal to the order due date

Which of the functional areas involved with APS are responsible for both providing valid data and for responding to APS outputs? Select all that apply.

(a) Planning
(b) Purchasing
(c) Inventory
(d) Order entry
(e) Shop floor
(f) Engineering

Refer to Appendix B: Check your understanding answers on page 110 for answers to the check your understanding questions.



Lesson 2: APS parameters and setup

Estimated time

0.5 hours

Learning objectives

After completing this lesson, you will be able to identify important APS parameters and setup requirements. In this lesson, you will:

- Describe key parameters used in planning and scheduling.
- Identify the different planning modes available with APS.

Topics

- Planning parameters
- Item settings affecting APS
- Planning modes

Planning parameters

There are numerous parameters and settings which affect APS and how it operates. In this lesson we will introduce several of the most important settings and the choices you'll need to consider. The specific selections you make must align with your business processes and your organization's needs.

In most cases, the best approach is to use recommended or default settings to start, and adjust as needed once after running APS for a period of time.

You'll want to work with an experienced consultant to develop a good process for planning with APS and then select the set of planning parameters which support your process.

This form is now available in a new gen2 version.

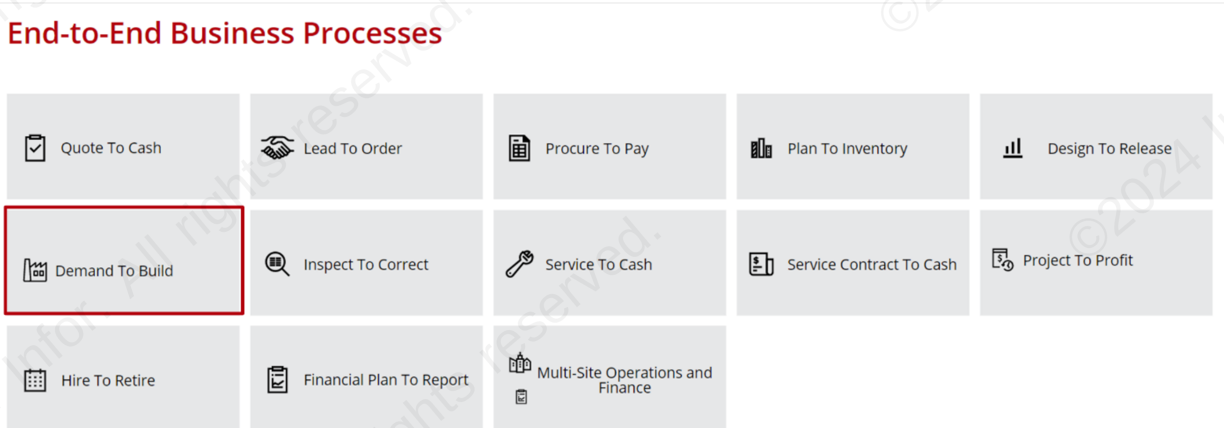
To learn more about gen2 forms, click here.

It's essential that you set up your planning parameters correctly. Review the material in this lesson carefully, and discuss with your consultant to determine the best settings for your situation. Remember, keep it simple to start and adjust as needed after you learn how the standard settings work in your situation.

Infor Implementation Accelerator (IA) for SyteLine/CloudSuite Industrial

As reviewed in the APS Overview and Concepts course, Infor Implementation Accelerators, or IAs, are a set of prescriptive, industry leading process flows for Infor CloudSuite applications. They deliver industry-specific leading practice content, including pre-defined configuration settings, process flows, and support assets.

Each CloudSuite IA is designed around the end-to-end business processes used by most organizations. The diagram below displays the full set of defined end-to-end business processes for CloudSuite Industrial. APS functions are included in the Demand To Build business process.



The Demand To Build business process encompasses all the steps involved in recording demand, planning, scheduling, and manufacturing to satisfy the recorded demand. Key capabilities include:

- Analyze demand

- Generate material plan
- Schedule shop floor
- Manufacture items to satisfy demand

As you can see from the diagram below, there are three sub-processes within Demand To Build. The Plan To Schedule sub-process is split into Procurement and Production. These are the primary process flows where APS is used.

Planning Parameters form: IA recommended settings

The Planning Parameters form is used to specify features and options used throughout the APS planning functions. The selections you make on this form will have a significant impact on the plans and outputs generated by APS.

The IA for SyteLine/CloudSuite Industrial provides recommended settings for each field on the Planning Parameters form. The full set of recommended settings can be found in Appendix C: Setting up APS on page 118. This appendix also includes reference material for other settings you may need to consider.

Some default settings are different from the settings recommended by the IA. It's a good idea to review all the settings on the Planning Parameters form and change selections as needed to match the IA configuration.

Be sure to discuss your organization's specific needs with an experienced consultant before using parameter settings other than those recommended by the IA.

Enforcing Get ATP/CTP

The Get ATP/CTP button can be used to help set realistic order promise dates on a real-time basis. This process allows you to test the viability of a demand and possible availability date against the current state of your APS plan.

Note: The button shown depends upon your planning mode. In Infinite APS mode, it is available-to-promise (ATP). In APS mode, it is capable-to-promise (CTP).

The ATP/CTP process returns a projected availability date for the demand. If this date is acceptable, you can allow the system to insert the demand incrementally into the actual plan.

To use the ATP/CTP calculated date on all customer order lines by default, select Calculate ATP/CTP for All CO Lines on the Planning Parameters form. In this case, the application will enforce Get ATP/CTP on all lines as they are entered. And you can decide to accept or reject the Get ATP/CTP date for each line.

Item settings affecting APS

Items are the key component of demands, and therefore of planning. Be sure to consider the following item factors in the context of APS. See Appendix C: Setting up APS on page 118 for additional details.

Items form - Planning menu option

The following fields and check boxes on the Items form - Planning menu option are important in APS functions.

Planner Code

This field identifies the planner responsible for this item. Many reports and forms can be filtered by or display the planner code.

Shrink Factor

Use this field to specify the percentage factor by which you want to increase the quantity of the planned job or order to compensate for expected loss prior to receiving an item to the stockroom. The system decreases the outstanding receipt of the planned job or order by the same amount.

Enter the percentage in decimal format. For instance, if you want the shrink factor to be 20%, type 0.2. The calculation for shrinkage is: Required quantity = desired quantity / (1 - shrink factor).

Example: you need 100 parts of Item A, and its shrink factor is 20%. The total units of item A to produce is determined by this formula: 100 / (1 - 0.2) = 125.

Therefore, enough components are planned to produce 125 units. Capacity for producing the 125 units will be reserved, and a plan for 100 units of Item A will be created.

The Shrink Factor field is linked to the Yield field on the item's current operations. When the yield is changed for one or more operations for a given item, that item's shrink factor is recalculated. When the shrink factor is changed, the yield on the last operation on the item's routing is recalculated.

Phantom Flag

Select this check box if the system should consider the item a phantom item. A phantom item is usually a sub-assembly that is not stocked, although you may sell it as a replacement part occasionally, and hence should not be part of the plan.

Lead times

APS uses lead times to approximate the time needed to acquire or receive purchased and transferred items. For manufactured items, APS plans using the times stated on the item's BOM; however, in special circumstances it will also use the item's lead times.

Use these fields to define lead time for APS Planning:

- Fixed Lead Time
- Variable Lead Time
- Expedited Fixed Lead Time
- Expedited Variable Lead Time
- Paper Work Lead Time

- Dock-to-Stock Lead Time

Note: Paper Work Lead Time and Dock-to-Stock Lead Time are also used by the MPS Processor, Material Planner Workbench Generation, and on the Order Action Report.

Purchased item lead time

For purchased items, you must enter the lead time values manually in the lead time fields. Use fixed lead time and possibly a dock-to-stock and paper work lead time; you will normally not use variable lead time.

In addition to calculating PLN release dates, the system uses purchased item lead time in these situations:

- When you cross-reference and create a PO from a job material. The system deducts the item's dock-to-stock lead time from the operation's start date, if not blank, or the job's start date if the operation's start date is blank.
- When you manually create a PO line or PO requisition line and a vendor contract exists for the item and the PO vendor, the system calculates the default due date by adding the item vendor record's lead time to the appropriate date.
  - PO line: Adds item vendor lead time to PO order date.
  - Blanket PO release: Adds item vendor lead time to the release date you entered.
  - PO requisition line: Adds item vendor lead time to the requisition date.
- To calculate the due date of safety stock planned orders (current date/time + lead time).

Manufactured item lead time

Normally, APS calculates the release date for planned manufacturing orders using the current operation times and any constraints caused by materials on the BOM or resource availability (when the required resources are on shift and any capacity constraints).

However, the system does use lead time to plan manufactured items in these situations:

- To plan the start date of a job when any of these conditions are true:
  - The item has no routing.
  - The MRP Item field is selected for the item on the Items form.
- To calculate the due date of safety stock planned orders (current date/time + lead time).
- To determine which routing and BOM to use when planning a requirement for an item.
  - The system decides which routings and BOMs to use through the indented bill of manufacturing by using lead time to estimate when they will be used, and then comparing those times to the effective dates for the possible routings and BOMs.

For manufactured items, you can enter the lead times manually or use the Lead Time Processor to generate the lead times from the current routing's operation times.

Expedited lead times

For special cases where you need to reduce the normal lead time, you can use the expedited lead time features. APS uses expedited lead time when the initial pull planning using standard lead time projects a date in the past.

You must select the Use Expedited Lead Time for Planning option on the Planning Parameters form to apply expedited lead times in APS.

You can apply expedited lead times for specific items or apply to all items. To apply on an individual item, specify a value in the Expedited Fixed Lead Time or Expedited Variable Lead Time field on the Items form. In this case, the expedited values replace the item's normal lead time values when needed, and override any global settings.

To apply on all items, specify a value in the Variable Lead Time Reduction (Hours) or Fixed Lead Time Reduction (Hours) on the Planning Parameters form. In this case, when APS identifies a need for expedited lead times, the standard lead time values on the Items form will be reduced by the number of hours in the expedited lead time reduction fields on the Planning Parameters form.

Items form - Advanced Planning menu option

Some additional important check boxes are found on the Items form - Advanced Planning menu option, as described below.

Accept Requirements and Pass Requirements

The Accept Requirements and Pass Requirements check boxes on the Additional Planning menu option of the Items form allow you to tell APS which items to plan and whether to pass demand down to component items.

Here is a description of how these fields work.

Check box	Purpose
Accept Requirements	When the Accept Requirements check box is selected, APS plans the item for parent demands. If Accept Requirements is cleared, APS does not plan the item for parent demands.
Pass Requirements	When the Pass Requirements check box is selected, APS will calculate requirements for the item's components. If the Pass Requirements check box is cleared, then no requirements are passed down to any of the item's materials. Typically, this check box is cleared for all purchased items to prevent APS from creating PLNs for the components of items that are purchased.

Safety stock and preservation levels

APS will create PLNs to help you maintain safety stock levels. APS will also help you protect near-term supply for near-term demands, reserving safety stock for orders inside an item's lead time.

Here are the fields used to set it up.

Form	Fields	Purpose
Item/Warehouse	Safety Stock	Sets the safety stock level you want to maintain
Items - Additional Planning menu option	Time Fence Rule Time Fence Value	Sets length of time to preserve safety stock quantities for near-term demands
	Pull-Up SS Rule Pull-Up SS Value	Allows you to set due dates earlier than the standard due date for safety stock replenishments **Note:** These are advanced parameters that you will want to discuss with your consultant.
Planning Parameters - Advanced APS tab	Plan Safety Stock Before Forecasts	Prevents forecasts from consuming safety stock quantities **Note:** This is an advanced parameter that you will want to discuss with your consultant.
	Safety Stock Sequence Rule	Determines which safety stock orders are highest priority to plan **Note:** This is an advanced parameter that you will want to discuss with your consultant.

Planned Orders Consolidation

The Planned Orders Consolidation parameter is used to consolidate planned orders for purchased items, manufactured items, or both. Here are the fields used to set it up.

Form	Field	Purpose
Planning Parameters: APS tab	Planned Orders Consolidation: Manufactured Items Planned Orders Consolidation: Purchased/Transferred Items	The values selected in these fields determine whether PLNs will be consolidated, and if so, the consolidation period to apply.
Items: Additional Planning menu option	Days Supply	When the Days Supply method is selected on the Planning Parameters form, the value in this field is the number of manufacturing days the system looks into the future for consolidating planned orders.

Days Supply

If you have the Planned Orders Consolidation fields on the Planning Parameters form set to "Days Supply," then this field is used to determine the set of PLNs that will be consolidated into one order.

The Days Supply field specifies how many days to search to combine PLNs into one PLN. This works much like the 30-day and 90-day bucket fields on the Planning Parameters screen.

For example, suppose APS creates PLNs for FA-10000 as shown in the table below.

Order #	PLN due date	Quantity
1	May 1	10
2	May 1	15
3	May 2	7
4	May 5	20
5	May 9	15
6	May 10	5

If you have set Days Supply to 5, then before the planning activity is done, it will combine all the PLNs for FA-10000 in five day increments, and you will see two PLNs as shown below.

Combine PLNs with due dates between	Consolidated order quantity	Consolidated order due date
May 1 - May 5	52	May 1
May 6 - May 10	20	May 9

Notice the due date of the consolidated PLN is the due date of the earliest PLN in the set of orders consolidated.

If you have selected Days Supply for order consolidation on the Planning Parameters form, you can still turn order consolidation off for any individual item by setting Days Supply for that item to 0.

If any of the consolidation parameters will affect manufactured items, Infor recommends that you enable the Plan Days Supply parameter on the Planning Parameters form and the Planned Mfg Supply Switching parameter on the Items form.

Order Minimum

Use this field to specify the minimum amount of this item to order. When the system creates a PLN for this item, it is for a quantity no less than the amount in this field. The system uses the Order Minimum in conjunction with the Order Multiple value. If the demand quantity is less than the Order Minimum value, the system creates the PLN for the Order Minimum value. If the Order Minimum value is less than the demand quantity, the system adds order multiples to the Order Minimum value until the PLN quantity is greater than or equal to the demand quantity.

Other demands can use any leftover quantities created by Order Minimum/Order Multiple calculations.

Order Multiple

Use this field to specify the multiple in which quantities of this item should be planned. The system uses the value in the Order Multiple field in conjunction with the Order Minimum value. If the demand quantity is less than the Order Minimum value, the system creates the PLN for the Order Minimum value. If the Order Minimum value is less than the demand quantity, the system adds Order Multiples to the Order Minimum value until the PLN quantity is greater than or equal to the demand quantity.

Other demands can use any leftover quantities created by Order Minimum/Order Multiple calculations.

Order Maximum

This field specifies the maximum amount of this manufactured end item to build. When planning the end item, the system breaks the item quantity into multiple loads of the Maximum Size value specified in this parameter.

For example, if the item's Order Maximum is 50, and an order for 100 is entered, the system plans two demands, each with a quantity of 50. If you are not using planned order consolidation, the system creates two PLNs (each for 50). Applying an Order Maximum value to an item allows you to load a demand across many resources to improve efficiency and speed.

The last load will contain the remainder of the original quantity, adjusted by any Order Minimum or Order Multiple calculations. For example, if the Order Maximum value is 1000, and the order is for a quantity of 100,002, the system plans the order as if it were for 100 line items of 1000 items each and one line item of two items.

The system applies Order Maximum before allocating planned supplies or on-hand inventory, and before applying the Shrink Factor, Order Minimum, and Order Multiple. If Order Maximum is less than the Order Minimum value, Order Maximum is ignored.

Specifying an Order Maximum value allows APS to split a demand into a number of loads. For example, if you setan Order Maximum value of 20 and receive a CO for 100, APS will create five PLNs with a quantity of 20 to supply the CO line.

APS will create multiple PLNs based on this field. If you firm these PLNs, they will firm up into separate orders.

These order parts or "lots" are maintained by APS. For example, when you view the order in the Gantt display, you will be able to see the lots the order has been broken into. The lots created by applying an Order Maximum value have nothing to do with lot tracking.

Planning modes

One of the fundamental decisions you must make when setting up APS is whether to run the system in Infinite APS or APS Mode. The difference between these modes of planning is in the resource capacity the system considers when generating a plan.

What you decide can have a dramatic effect on the projected shipping dates, and, therefore, what dates you promise your customers. It will also affect the dates APS tells you to release your supply orders since those are tied to the projected date.

Infinite APS assumes infinite resource capacity, while APS Mode constrains the plan realistically based on availability of resources.

Planning mode	Description
APS Mode	APS Mode plans all requirements for one demand before planning the next demand, based on order priorities and due dates. It considers whether resources (crew, machines, and fixtures) are on shift and not busy working on other demands. Resources can work on only one task at a time. That is, if drillpress_ABC is working on Operation 10 of Job 123 at 10:00 a.m., drillpress_ABC cannot also work on Operation 10 of Job 456 at 10:00 a.m. APS generates PLNs and exception messages, and reports detailed resource usage. On forms such as Customer Order Lines and Job Orders, the Get ATP/CTP buttons will be labeled Get CTP.
Infinite APS	Infinite APS plans all requirements for one demand before planning the next demand, based on order priorities and due dates. However, unlike APS Mode, it assumes your resources (crew, machines, and fixtures) can work on an unlimited number of tasks while in a working shift period. The system limits a resource's production only by the available on-shift time. Infinite APS generates PLNs and exception messages, and reports detailed resource usage. The Get ATP/CTP buttons will be labeled Get ATP.

Factors to consider when selecting the planning mode

As reviewed in the APS Overview and Concepts course, it's critical to have all of your planning policies and procedures working together in alignment with your business process. Determining whether to schedule with finite or infinite capacity is one of the process decisions you must make.

For most customers, Infor recommends using infinite capacity. However, some organizations really do have inflexible capacity, and in these cases finite capacity scheduling may be a better option.

How can you know which is the right choice for you?

We've found the factors below are good indicators. The more of these that apply in your situation, the less flexible your actual capacity might be, and the more you might be a candidate for finite capacity scheduling.

- Single, stable, chronic bottleneck
- Utilization is regularly above 80%

- Little room to schedule overtime
- Difficult to outsource work
- Direct labor not cross-trained
- Cycle time in weeks
- Must promise customer orders in real time, in an environment with the other factors listed

Please note that the factors above are not determinative. Even if you deal with many of them, you may still find that infinite capacity scheduling is the best option.

In most cases, Infor recommends using Infinite APS, but this selection must be reviewed for your specific requirements. Be sure to work with an experienced consultant to determine the best solution for your organization.

Exercise 2.1: Set Infinite APS

In this exercise, you will set the planning mode to Infinite APS.

Before you begin:

- Ensure you are logged in to SyteLine.

Exercise steps

Additional learners

If you have multiple learners using your training environment only one person can perform the steps to update the APS planning mode. We recommend you do the exercise steps, and if the mode has already been updated take the time to review the settings.

1. Click the **Open** icon on the toolbar. The **Select Form** dialog box opens.
2. Type *infinite* in the **Filter** field.
3. Click the **Infinite APS Mode** list item.
4. Click the **OK** button. The **Infinite APS Mode** form opens.
5. Click to select the **Use Infinite APS Planning** check box.
6. Click the **Save** icon.
7. Click the **X** on the **Infinite APS Mode** form.

Check your understanding

Which form is used to specify APS features and options?

(a) APS Options form
(b) Scheduling Parameters form
(c) APS Sites and Management form
(d) Planning Parameters form

Which of the following statements about lead times are true? Select all that apply.

(a) Lead times for purchased parts are normally entered on the Items form.
(b) Lead times for manufactured parts are normally entered on the Items form.
(c) Lead time values on the BOM are used to calculate precise projected dates for all items.
(d) Expedited lead times are specified only on an item-by-item basis.
(e) Expedited lead times can be applied on a global basis or an individual item basis.

When might APS load a resource at over 100% of capacity?

(a) Never
(b) When the first pull pass of the planning process fails
(c) When the system is set for APS Mode
(d) When the system is set for Infinite APS

Refer to Appendix B: Check your understanding answers on page 110 for answers to the check your understanding questions.

Lesson 3: Modeling the site for planning

Estimated time

1 hour

Learning objectives

After completing this lesson, you will be able explain to how to model the site to use APS. In this lesson, you will:

- Describe how resources are used in APS.
- Identify the different types of records used in APS.
- Explain the importance of an item's BOM.

Topics

- Overview
- Resources and resource groups
- Course scenario
- Work centers
- Items
- Routing
- Materials
- Customer orders

Overview

When you model the environment, you're telling APS about the:

- Resources you have
- Items you make, their components, and how to plan them
- BOMs you use for manufactured items
- Global settings you want to use when planning

Before you begin to set up the data for your model, make sure to consider the following topics.

Your current situation

Consider these factors for your current situation:

- What are your key planning and scheduling constraints?
- What lead time and BOM data do you have?

Item and resource settings

Some considerations for your item and resource settings are:

- Which items will you plan with APS?
- Which method will you use to plan each of those items?
- What are the lead times for purchased and transferred items?
- Which shop floor resources need scheduling?
- What are the BOMs for your manufactured items?
- Which planning adjustments do you want to make for each item?

Site settings

Site-wide settings to consider include:

- Will you use APS or the Scheduler to create the shop floor dispatch schedule?
- Which planning adjustments do you want to make for the site?
- How many What-If Analysis alternatives will you have?
- If you are multi-site, are you going to plan at the site or global level?

For APS to work effectively for your company, you need to make sure your parameters and configuration are set up correctly. Infor strongly recommends that you do not attempt to configure your system without the help of an experienced consultant.

Resources and resource groups

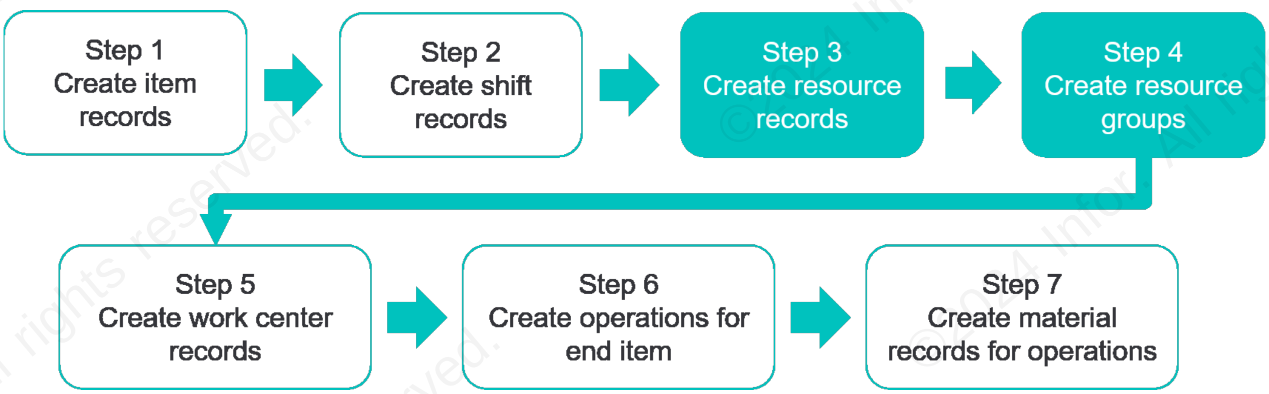
A resource is a non-specific person, machine, or other tool used in at least one step (operation) of the manufacturing process. Examples of a resource are:

- Machine
- Crew
- Fixture
- Space
- Tool

Each operation specifies one or more resource groups. A resource group is a collection of similar resources that can perform the operation. The system selects one or more resources from the assigned resource group to perform the operation, depending on each resource's availability and on rules you define.

The member resources may be working different shifts, have different current setups, or have different processing times. You can define a resource as a member of more than one resource group. Also, all the members in a resource group need not be from the same resource type. Each resource should be a member of at least one resource group.

When planning and scheduling demands, SyteLine needs an accurate representation of the availability and capacity of your resources. Therefore, resources and resource groups are both needed for APS, along with other elements of the item's BOM as shown below.



The following components comprise the system resource model:

- Resources
- Resource groups
- Scheduling shifts
- Shift exceptions
- Holidays

Modeling labor resources

When APS allocates a resource to an operation, it assumes that resource will perform the work from start to finish. This means when the resource goes off shift, APS plans for that resource to continue when it comes back on shift.

This planning method makes sense for machine resources. However, when you're scheduling labor resources and you have multiple shifts, you probably don't want APS to schedule a job to be worked on only by the labor resource in one shift—you want the resources working subsequent shifts to work on that same job as well.

For example, consider an assembly area that runs 16 hours per day with two labor resources as shown below. The first labor resource works shift 1; the second works shift 2.

	Day 1			Day 2		
Assembly	Shift 1	Shift 2	No Shift	Shift 1	Shift 2	No Shift
Labor 1						
Labor 2						

Let's say you get some work that will take 16 hours to finish and it's due at the end of day 1. You would want APS to schedule as shown below.

	Day 1			Day 2		
Assembly	Shift 1	Shift 2	No Shift	Shift 1	Shift 2	No Shift
Labor 1	Load					
Labor 2		Load				

But APS won't schedule it that way with the above resource modeling. It will load the work on Labor 1 and not reassign the work when Labor 1 goes off shift as shown below.

	Day 1			Day 2		
Assembly	Shift 1	Shift 2	No Shift	Shift 1	Shift 2	No Shift
Labor 1	Load			Load		
Labor 2						

To ensure that APS schedules the labor as you desire, simply model a single labor resource at the assembly area that works both shifts, as shown below.

	Day 1			Day 2		
Assembly	Shift 1	Shift 2	No Shift	Shift 1	Shift 2	No Shift
Labor 1	Load	Load				

Labor resources need to be thought of as available person hours per day rather than as individuals.

What if you don't have the same number of labor resources on first shift as you do on second? For example, what if you have two people on first shift and one person on second?

Again, you want to model available person hours per day. Here are two ways you could do it:

- Option 1
  - Resource A: Available 1.5 shifts
  - Resource B: Available 1.5 shifts
- Option 2
  - Resource A: Available 2 shifts
  - Resource B: Available 1 shift

Both options reflect how much capacity you have.

Please note that Option 2 is less constraining than Option 1, because it will allow you to use up to two full shifts of linear time each day. If you're running APS in infinite mode and have made Resource A the first resource in the sequence for the resource group, using Option 2 can reduce the number of times a projected date pushes out because you don't have enough on-shift time.

With either option, if you go over capacity, you can see it with the APS alerts and address the issue.

Modeling specific and alternate resources

You can set up your resource model to use a single individual resource on an operation, or multiple alternate resources.

How to model specific resources

To assign one specific resource to work on an operation, create a resource group that contains only that resource.

For example, assume you have five lathes. One of them is new and is the only machine that you can use on a subset of the items you manufacture. You can create these resource groups:

Resource group	Resources
Lathes	• Old lathe 1 • Old lathe 2 • Old lathe 3 • Old lathe 4 • New lathe 1
New Lathe	• New lathe 1

Then on the Operations form for the items that require the new lathe, specify the New Lathe resource group. APS will only load those items on the new lathe.

How to model alternate resources

The resource model allows you to specify alternate resources for a routing by using resource groups and resource group member selection rules.

For example, suppose there is a lathe operation for a jet engine ring that normally requires New lathe 1. If that lathe is not available, the job can alternatively be run using Old lathe 1 or Old lathe 3. However, you always want to use New lathe 1 if it is available.

Here is how you could set up the resource group that is assigned to the operation:

Resource group	Resources
Jet engine ring lathes	• New lathe 1 • Old lathe 1 • Old lathe 3

On the General menu option of the Resource Groups form, an allocation rule can be selected from a drop-down list. Use the Select In Sequence allocation rule to specify that the resource group's resources are to be selected in the same order as they are listed on the Resources menu option.

For operations that are assigned this resource group, if New lathe 1 is available, it is loaded. If not, then the next available resource in the list is loaded to perform the operation.

Because resources can be members of many groups, you could create many such resource groups in which New lathe 1 is a member or the primary resource.

Course scenario

In the rest of this course, we will use a specific business scenario to illustrate the concepts and tasks of using APS. First, we will model the site using this scenario, and then explore options and outcomes based on the scenario.

Scenario: Summary

Our demonstration corporation, Progressive Cycles, is constantly adding new products. The newest addition to the product line is a Deluxe Child Bicycle Trailer (DCBT-100). In this lesson, we will set up the components needed to manufacture the trailer and model the relationships between them. This includes:

- Resources
- Resource groups
- Work centers
- Items
- BOM

To start with, we will create resource and resource group records for our scenario.

Exercise 3.1: Create resource records

In this exercise, you will create three resource records and tie them to the appropriate shift.

Before you begin:

- Ensure you are logged in to SyteLine.

Exercise steps

Part 1: Create welding resource

1. Click the **Open** icon on the toolbar. The **Select Form** dialog box opens.
2. Type *resources* in the **Filter** field.
3. Click the **web.Resources** list item.
   Note: The web.forms will be listed under the Name column.
4. Click the **OK** button. The **Resources** form opens.
5. Click the **New** icon.
6. Type *[your initials]-DCBT-MR1* in the **Resource** field.
7. Type *Welding* in the **Description** field.

8. Verify **Machine** is displayed in the **Resource Type** field.
9. Click the **Shifts** menu option.
10. Click the **Shift ID#1** drop-down arrow.
11. Click the **1shift** list item.
12. Click the **Shift ID#2** drop-down arrow.
13. Click the **2shift** list item.

Part 2: Create painting resource

1. Click the **New** icon.
2. Type *[your initials]-DCBT-MR2* in the **Resource** field.
3. Type *Painting* in the **Description** field.
4. Verify **Machine** is displayed in the **Resource Type** field.
5. Click the **Shift ID#1** drop-down arrow.
6. Click the **1shift** list item.
7. Click the **Shift ID#2** drop-down arrow.
8. Click the **2shift** list item.

Part 3: Create labor resource

1. Click the **New** icon.
2. Type *[your initials]-DCBT-LR1* in the **Resource** field.
3. Type *Assembly* in the **Resource Description** field.
4. Click the **Resource Type** drop-down arrow.
5. Click the **Labor** list item.
6. Click the **Shift ID#1** drop-down arrow.
7. Click the **1shift** list item.
8. Click the **Shift ID#2** drop-down arrow.
9. Click the **2shift** list item.
10. Click the **Save** icon.
11. Click the **X** on the **Resources** form.

Exercise 3.2: Create resource groups

In this exercise, you will create resource groups and tie them to the resources you just created.

Before you begin:

- Ensure you have completed Exercise 3.1 because it provides data or configurations for this exercise.
- Ensure you are logged in to SyteLine.

Exercise steps

Part 1: Create welding resource group

1. Click the **Open** icon on the toolbar. The **Select Form** dialog box opens.
2. Type *resource groups* in the **Filter** field.
3. Click the **web.ResourceGroups** list item.
4. Click the **OK** button. The **Resource Groups** form opens.
5. Click the **New** icon.
6. Type *[your initials]-DCBT-RGW* in the **Group** field.
7. Type *Welding* in the **Description** field.
8. Verify **Machine** is displayed in the **Type** field.
9. Click the **Resources** menu option.
10. Click the **Resource** field in **row 1** of the grid.
11. Type *[your initials]* in the **Resource** field.
12. Click the **Resource** drop-down arrow.
13. Click the **[your initials]-DCBT-MR1** list item.
14. Click the **Save** icon.
15. Click the **Description** field to prepare to add another resource group.
    Note: If you miss this step, you will create a new record in the Resources grid instead of a new resource group record.

Part 2: Create painting resource group

1. Click the **New** icon.
2. Type *[your initials]-DCBT-RGP* in the **Group** field.
3. Type *Painting* in the **Description** field.
4. Verify **Machine** is displayed in the **Type** field.
5. Click the **Resource** field in **row 1** of the grid.
6. Type *[your initials]* in the **Resource** field.
7. Click the **Resource** drop-down arrow.

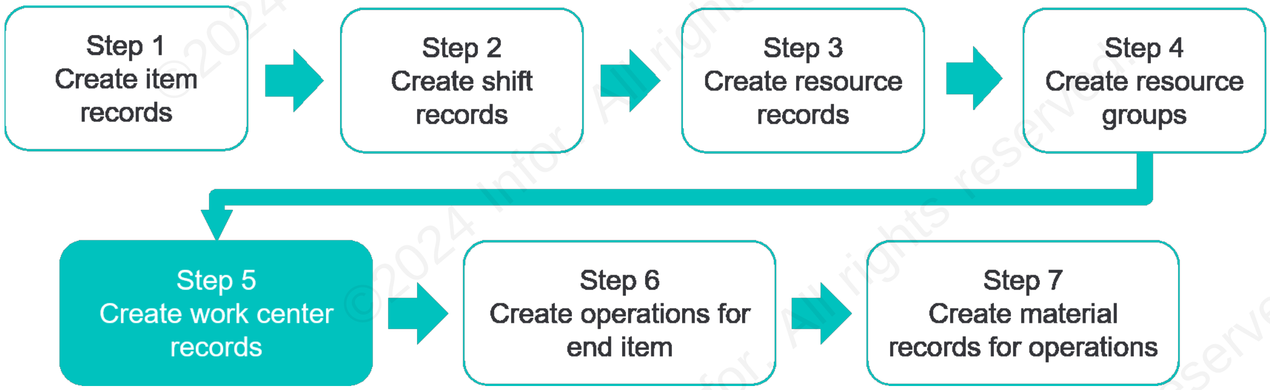
8. Click the **[your initials]-DCBT-MR2** list item.
9. Click the **Save** icon.
10. Click the **Description** field to prepare to add another resource group.

Part 3: Create assembly resource group

1. Click the **New** icon.
2. Type *[your initials]-DCBT-RGA* in the **Group** field.
3. Type *Assembly* in the **Description** field.
4. Click the **Type** drop-down arrow.
5. Click the **Labor** list item.
6. Click the **Resource** field in **row 1** of the grid.
7. Type *[your initials*] in the **Resource** field.
8. Click the **Resource** drop-down arrow.
9. Click the **[your initials]-CBT-LR1** list item.
10. Click the **Save** icon.
11. Click the **X** on the **Resource Groups** form.

Work centers

The diagram below illustrates the components of a BOM. As you can see, work centers are needed for each BOM.



Effect of work centers on planning and scheduling

Work centers are not directly used in planning and scheduling activities. However, the data in some of the fields on the Work Centers form generates default values for some fields on the Operations form. Most of these fields are related to costing, but a few of them affect planning and scheduling. These are:

- Schedule Driver (defaults from resource group)
- Efficiency (affects cost and scheduling)
- Queue Hours (wait time prior to operation)
- Finish Hours (wait time after operation)
- Resource Groups (scheduling)

All work center defaults may be overwritten at the operation level. You do not have to specify any field on a work center record other than the Department field; other fields are used for convenience when setting up operations.

Scenario: Work centers

The next step in setting up the manufacturing components is to create work centers for our new product. Work centers are used to track the production costs that planning and scheduling activities generate.

Exercise 3.3: Create work centers

In this exercise, you will create work centers and tie them to the three resource groups you created.

Before you begin:

- Ensure you have completed Exercise 3.2 because it provides data or configurations for this exercise.
- Ensure you are logged in to SyteLine.

Exercise steps

Part 1: Create the welding work center

1. Click the **Open** icon on the toolbar. The **Select Form** dialog box opens.
2. Type *work centers* in the **Filter** field.
3. Click the **web.WorkCenters** list item.
4. Click the **OK** button. The **Work Centers** form opens.
5. Click the **New** icon.
6. Type *[your initials]-DWW* in the **Work Center** field.
7. Type *Welding* in the **Name** field.
8. Click the **Department** drop-down arrow.
9. Click the **200** list item.
   Note: The department selected will not have an effect on scheduling. Any department can be selected.
10. Click the **Resource Groups** menu option.
11. Click the **Group** field in **row 1** of the grid.
12. Type *[your initials]* in the **Group** field.
13. Click the **Group** drop-down arrow.
14. Click the **[your initials]-DCBT-RGW** list item.
15. Click the **Save** icon.
16. Click the **Scheduling** menu option.

Part 2: Create the painting work center

1. Click the **New** icon.
2. Type *[your initials]-DWP* in the **Work Center** field.
3. Type *Painting* in the **Name** field.
4. Click the **Department** drop-down arrow.
5. Click the **200** list item.
6. Click the **Resource Groups** menu option.
7. Click the **Group** field in **row 1** of the grid.

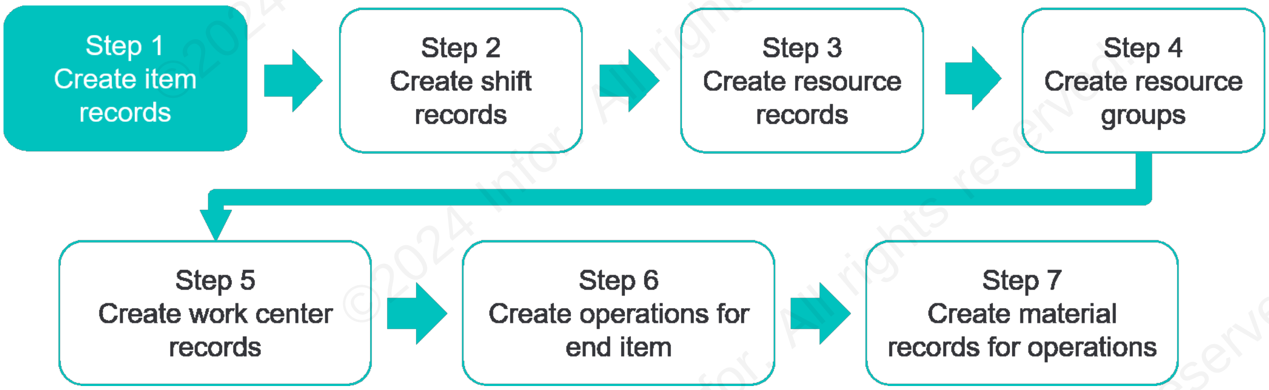
8. Type *[your initials]* in the **Group** field.
9. Click the **Group** drop-down arrow.
10. Click the **[your initials]-DCBT-RGP** list item.
11. Click the **Save** icon.
12. Click the **Scheduling** menu option.

Part 3: Create the assembly work center

1. Click the **New** icon.
2. Type *[your initials]-DWA* in the **Work Center** field.
3. Type *Assembly* in the **Name** field.
4. Click the **Department** drop-down arrow.
5. Click the **100** list item.
6. Click the **Resource Groups** menu option.
7. Click the **Group** field in **row 1** of the grid.
8. Type *[your initials]* in the **Group** field.
9. Click the **Group** drop-down arrow.
10. Click the **[your initials]-DCBT-RGA** list item.
11. Click the **Save** icon.
12. Click the **X** on the **Work Centers** form.

Items

An item is something you have bought, manufactured, and/or sold. Creating item records is the first step in setting up your BOM:



The information on the Items form is used extensively throughout the system, and provides critical data for the BOM and for the APS planning and scheduling functions.

Item fields specific to BOMs

The following fields are those on the Items form, Detail menu option that are useful in a BOM.

U/M

U/M specifies the item's unit of measure.

Type

Each item is categorized by the Type field, as described below.

Type	Description
Material	Items manufactured or consumed in the manufacturing process
Fixture	Devices that support materials or tools during the machining process
Tool	Instruments used by a machine to perform an operation
Other	Non-tangible cost elements on a BOM such as engineering fees or outside process costs

Source

The Source field identifies how the item is added to inventory:

- Purchased
- Manufactured
- Transferred

The value you select in the Source field does not dictate that the item can only be purchased, manufactured, or transferred. Rather, your selection tells the system what to use as a default value.

Alternate Item

The Alternate Item field references a substitutable or a replacement part. It is used for reference only.

Low Level

The Low Level field specifies the low-level code, which indicates the lowest level of the item in any current, job, or production BOM. A low-level code of 0 indicates the item is an end item (finished goods) and is not a subcomponent of another item. A low-level code greater than zero indicates the item is a component of another item. The system supports up to 20 levels of low-level coding.

Stocked

This field has nothing to do with whether an item is normally stocked or not. When you select this field, you tell the system you want to use the dynamic allocation (pegging) feature of APS or MRP when planning this item.

Normally, whether you make or assemble this item to order or to stock, this field should be selected. When you clear the field, you are telling the system you want to manually create hard-allocated supplies for demands for this item and manually maintain all the changes in the related dates if the demand dates change.

When you're planning with APS or MRP, you normally want those planning systems to do such work for you, so you would select this field. When you clear the Stocked field, the Reference field on the Customer Order Lines form and Job Materials form default to a type other than Inventory so you can use the manual sourcing features.

The Stocked field and the Source field on the Items form work in conjunction to create default references throughout the system as shown below.

If Stocked check box status is:	And Item Source value is:	Then Reference field value on Current Materials form is:
Selected	Purchased	Inventory
	Manufactured	Inventory
	Transferred	Inventory
Cleared	Purchased	Purchase Order or Purchase Order Requisition
	Manufactured	Job
	Transferred	Transfer

Scenario: Items

The next step in our scenario setup is to create the new item records for the DCBT product. The Items form is used to maintain a list of all items bought, manufactured, and/or sold. This form contains such values as description, unit of measure, and lead times for an item. Finished goods, sub-assemblies, raw materials, tooling, fixtures, and even outside services should have item records.

In our scenario, we need two item records, one for the finished trailer product and one for the wheels, a purchased product used in assembling the trailer.

Exercise 3.4: Add items

In this exercise, you will continue building the deluxe child bicycle trailer scenario by creating item records for the finished product and for the wheels used in assembly.

Before you begin:

- Ensure you have completed Exercise 3.3 because it provides data or configurations for this exercise.
- Ensure you are logged in to SyteLine.

Exercise steps

Part 1: Create the deluxe child bicycle trailer item

1. Click the **Open** icon on the toolbar. The **Select Form** dialog box opens.
2. Type *items* in the **Filter** field.
3. Click the **web.Items** list item.
4. Click the **OK** button. The **Items** form opens.
5. Click the **New** icon.
6. Type *[your initials]-DCBT-100* in the **Item** field.
7. Type *Deluxe Child Bicycle Trailer* in the **Description** field.
8. Click the **U/M** drop-down arrow.
9. Click the **EA** list item.
10. Click the **Product Code** drop-down arrow.
11. Click the **FG-100** list item.
12. Click to select the **Stocked** check box in the **Miscellaneous** section.
13. Click the **Planning** menu option.
14. Type *[your initials]* in the **Planner Code** field.
15. Click the **Sales** menu option.
16. Type *4* in the **Std Due Period** field.
17. Click the **Save** icon.

Part 2: Add pricing

These steps provide information needed when entering customer order lines. Without pricing data, you will get error messages when adding order lines.

1. Click the **Orders** menu drop-down arrow.
2. Click the **Pricing** menu option. A dialog box opens with the message, “Not one Item Price exists for Item that has [Item: [your initials]-DCBT-100]”.
3. Click the **OK** button. The **Item Pricing (Linked)** form opens.
4. Type *125* in the **Unit Price 1** field.
5. Click the **Save** icon.
6. Click the **X** on the **Item Pricing (Linked)** form. The **Items** form displays.

Part 3: Create the item record for the purchased part

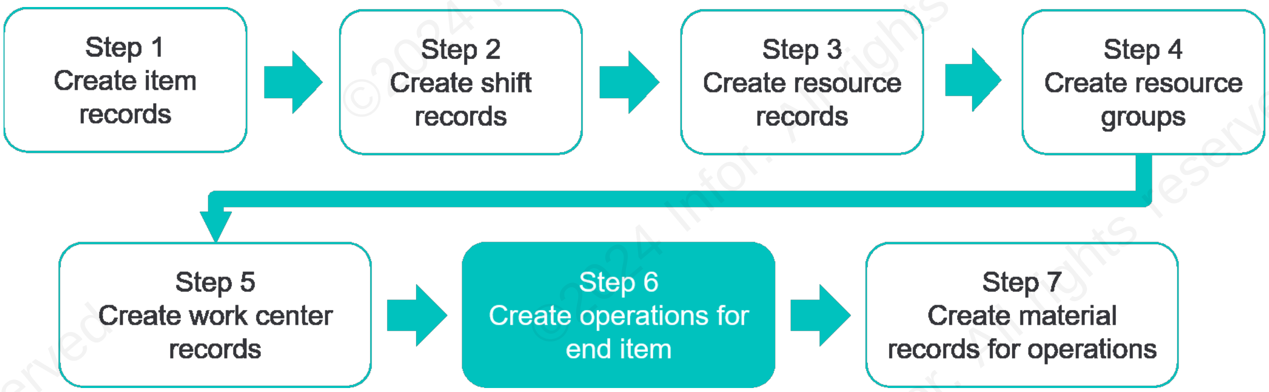
1. Click the **Items** menu drop-down arrow.
2. Click the **Detail** menu option.
3. Click the **New** icon.
4. Type *[your initials]-DCBTW-100* in the **Item** field.
5. Type *Wheels, Primary* in the **Description** field.
6. Click the **U/M** drop-down arrow.
7. Click the **EA** list item.
8. Click the **Source** drop-down arrow.
9. Click the **Purchased** list item.
10. Click the **Product Code** drop-down arrow.
11. Click the **PP** list item.
12. Click to select the **Stocked** check box in the **Miscellaneous** section.
13. Click the **Planning** menu option.
14. Type *[your initials]* in the **Planner Code** field.
15. Type *5* in the **Fixed** field under the **Lead Time (Days)** section.
16. Click the **Save** icon.
17. Click the **X** on the **Items** form.

Routing

Once resources, resource groups, and work centers are created, routings are the next step in creating your BOM. A routing is a set of operations and does not include material. It is required to have at least one operation before you can add any material to the BOM.

Creating operation records for an item

An operation record is a key component of the BOM.. It specifies the times, resources required, and costs of a manufacturing process step.



When creating or modifying operation information, you can use two forms:

- Engineering Workbench (for all BOM types)
- Operations form for the type of BOM you want to use

Current Operations – Details menu option

Details about the fields on the Current Operations form – Details menu option are described in the following tables.

Parent Item section

Field	Description
Revision	The revision number of the item, which helps you track which revision (authorized through Engineering Change Notices) the transaction affects. On some forms, this field is not enabled unless an item is selected.

Scheduling section

All times should be entered as hours in decimal format. For example, an expected setup time of 45 minutes (45/60) is entered as 0.75.

Field/check box/radio button	Description
Use Fixed Schedule	Select the Use Fixed Schedule check box to set a fixed time for this operation regardless of lot size. Selecting this check box also updates the Scheduler Rule on the Resources tab.
Fixed Sched Hours	If the Use Fixed Schedule check box is selected, enter the expected total amount of time for the operation in this field. When you use fixed hours, APS and the Scheduler ignore the values you have in the Hours per Piece and Setup fields. The processing time for the operation includes: • Fixed Sched Hours • Move hours • Queue hours • Finish hours
Machine Hours per Piece	Select the Machine Hours per Piece radio button to specify that the value that displays next to this field represents the hours of run time to produce one piece. For example, if the value is 2.0, and the job quantity is 10, the cycle time is 20 hours. If Sched Driver field = Machine, then enter a value in this field or in the Pieces per Machine Hour field.
Pieces per Machine Hour	Select the Pieces Per Machine Hour radio button to specify that the value field represents the pieces that can be produced in a single hour. For example, if the value is 2, two pieces can be produced every hour. If Sched Driver field = Machine, then enter a value in this field or in the Machine Hours per Piece field.
Labor Hours per Piece	Select the Labor Hours Per Piece radio button to specify that the value that displays next to this field represents an estimate of the hours of cycle time for one piece. For example, if the value is 2.0, and the job quantity is 10, the run time is 20 hours. If Sched Driver field = Labor, then enter a value in this field or in the Pieces per Labor Hour field.
Pieces per Labor Hour	Select the Pieces Per Labor Hour radio button to specify that the value field represents an estimate of the number of pieces that can be produced in a single hour. For example, if the value is 2, two pieces can be produced every hour. If Sched Driver field = Labor, then enter a value in this field or in the Labor Hours per Piece field.
Sched Driver	Use the Sched Driver field to specify whether machine or labor hours will be used for planning and scheduling the operation.

Field/check box/radio button	Description
Run Duration	This field displays the processing time (adjusted for efficiency) for the operation that APS and the Scheduler use to plan or schedule the resources for this operation. The system divides the Machine Hours Per Piece or Labor Hours per Piece value by the number of resources from the Schedule Driver resource group type and multiplies the value by (100/Efficiency).

Hours section

Field/check box	Description
Move	The Move field indicates the amount of time to schedule for movement of work in progress inventory from the previous operation to this operation
Queue Time	The Queue Time field indicates the Estimated amount of time you expect the item to wait before being processed at this operation. **Note:** APS applies this value only if the Infinite Resource Capacity After field is set to 0 (the queue hours are added to the time between the start of this operation and the start of the next operation).
Setup	The Setup field indicates the time to set up this operation. This value will be divided by the operation's Efficiency value.
Finish	The Finish field indicates the delay between completion of this operation and the start of the next operation in the routing.
Use Offset Hours	The Use Offset hours check box is selected to allow this operation to start before the previous operation completes. Then, in the Offset field, enter the number of hours this operation must wait to start after the previous operation starts. When the check box is cleared, this operation begins at the ending date and time of the previous operation.
Offset	The Offset field indicates the number of hours between the beginning of the previous operation and the beginning of this operation in the routing when the Use Offset Hours check box is selected. For example, suppose you have two operations, 10 and 20. If Operation 20 must start 30 minutes after Operation 10 starts, define Operation 20's Offset value as 0.5. If the Use Offset Hours check box is selected, and the Offset value is 0.0, then parallel operations are invoked. All operations which define Offset in succession are processed as parallel to the first operation. If there is move time for this operation, the system will not offset this operation.

Yield on job order operation affects the job's expected quantity, per the following formula:

Expected Quantity = Released Quantity * Yield

APS and MRP assume the expected quantity, not the released quantity, will be available for supply.

Current Operations – Resources menu option

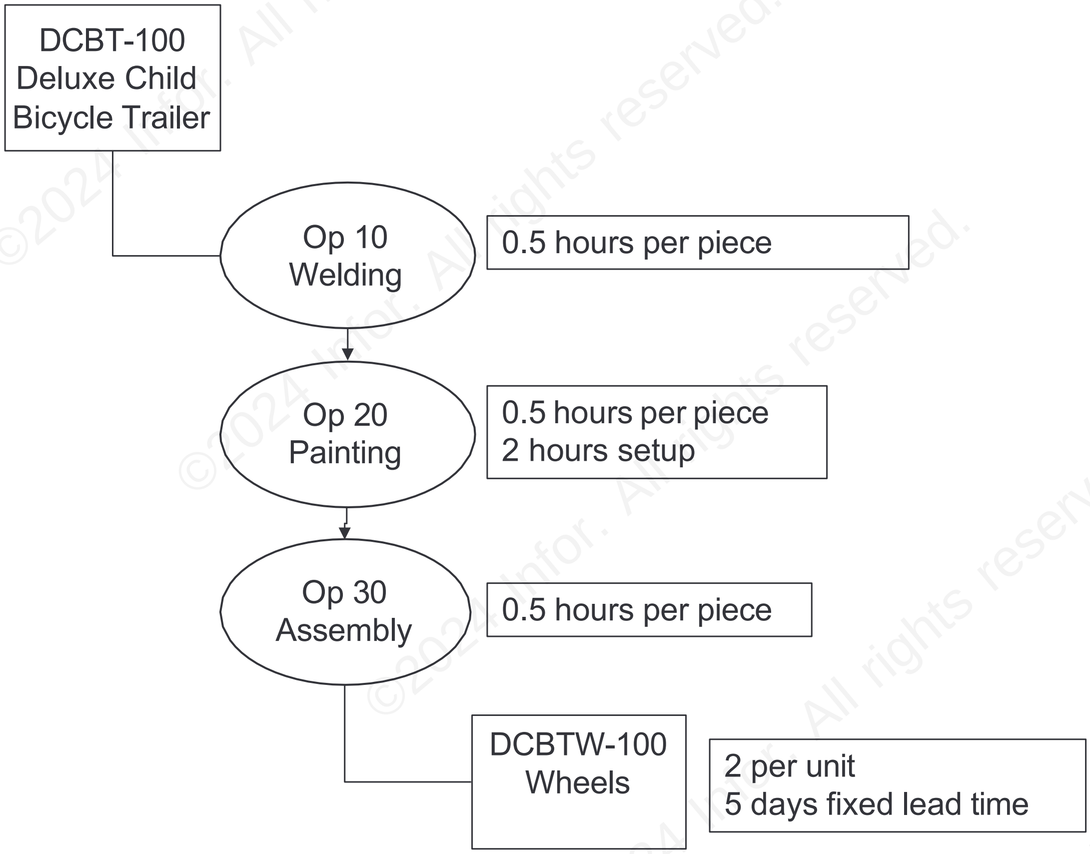
Details about key fields on the Current Operations form – Resources menu option are described in the following table.

Field	Description
Resources grid	
Resource Group	Select the resource group to work on this operation. You must specify at least one resource group, but not more than six, for each operation.
Quantity	Enter the number of resources from this resource group that should be allocated to a load. The default value is one. For example, define a quantity of three when the operation requires three crew members. Do not specify a quantity greater than the number of member resources defined for the resource group.
Resource Setup section	
Setup Resource Group	Select the resource group that is designated for the setup of this operation. The resource group you select must be allocated to this operation or must be allocated to a previous operation using the Hold action code; otherwise, an error occurs during scheduling. The resource allocated from this group is the resource the Scheduler uses to determine the setup time based on the criteria you defined (by item, by setup group, etc.). All resources required on the operation are allocated for the setup time. If you entered a setup time on the Standards tab, then you must select a setup resource group. **Note:** If you do not specify a setup resource group, this operation does not incur setup time.

Scenario: Operations

The next manufacturing data components needed for the deluxe child bicycle trailer BOM are the current operations records. These records define the quantity of resources required for each step of the manufacturing process.

Our scenario uses three operations: welding, painting, and assembly. The assembly operation requires the DCBTW-100 wheels materials, which we will add in the next section.



Exercise 3.5: Add current operations for DCBT-100

In this exercise, you will create new current operations records for DCBT-100.

Before you begin:

- Ensure you have completed Exercise 3.4 because it provides data or configurations for this exercise.
- Ensure you are logged in to SyteLine.

Exercise steps

Part 1: Create operation 10, welding

1. Click the **Open** icon on the toolbar. The **Select Form** dialog box opens.
2. Type *items* in the **Filter** field.
3. Click the **web.Items** list item.
4. Click the **OK** button. The **Items** form opens.

5. Click the **Filter in Place** icon.
6. Type *[your initials]-DCBT-100* in the **Item** field.
7. Click the **Filter in Place** icon. The **Item** record displays for your Deluxe Child Bicycle Trailer.
8. Click the **Production** menu drop-down arrow.
9. Click the **Current Operations** menu option. The **Current Operations (Linked)** form opens.
10. Type *[your initials]* in the **WC** field.
11. Click the **WC** drop-down arrow.
12. Click the **[your initials]-DWW** list item.
13. Type *.5* in the **Machine Hours per Piece** field.

Part 2: Create operation 20, painting

1. Click the **New** icon.
2. Type *[your initials]* in the **WC** field.
3. Click the **WC** drop-down arrow.
4. Click the **[your initials]-DWP** list item.
5. Type *.5* in the **Machine Hours per Piece** field.
6. Type *2* in the **Setup** field in the **Hours** section.
7. Click the **Resources** menu option.
8. Type *[your initials]* in the **Setup Resource Group** field in the **Resource Setup** section.
9. Click the **Setup Resource Group** drop-down arrow.
10. Click the **[your initials]-DCBT-RGP** list item.
    Note: Setup will not be applied unless you select a value here.

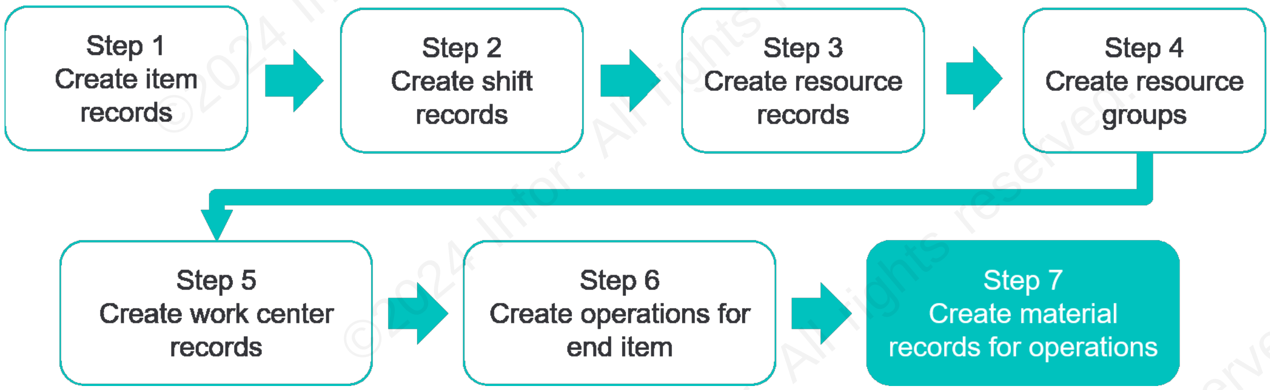
Part 3: Create operation 30, assembly

1. Click the **Details** menu option.
   Note: If you miss this step, you will create a new record in the Resources grid instead of a new current operations record
2. Click the **New** icon.
3. Type *[your initials*] in the **WC** field.
4. Click the **WC** drop-down arrow.
5. Click the **[your initials]-DWA** list item.
6. Type *.5* in the **Labor Hours per Piece** field.
7. Verify **Labor** is displayed in the **Sched Driver** field.
8. Click the **Save** icon.

9. Click the **Window** drop-down arrow on the menu bar.
10. Click the **Close All** list item.

Materials

The final step in creating your BOM is to add materials to operations.



A material specifies the items required to perform a given operation.

When creating or modifying material information, you can use two forms:

- Engineering Workbench (for all BOM types)
- Materials form for the BOM type you want to work with

In this course, we do not go into the details about all the costing fields for a materials record. Instead, we will simply:

- Create a material record.
- Fill in the quantity per fields.

Current Materials – Materials menu option

Details about the fields, check boxes, and radio buttons on the Current Materials form – Materials menu option are described in the table below.

Field/check box/radio button	Description
Material	Material is a required field. Select the material to be added to the operation from the drop-down list. This can be an inventoried or non-inventoried item. The system applies cost per unit for inventoried items. Non-inventoried items might be supply items which will not directly affect costing of the job. These charges could be captured in the overhead accounts.
Quantity	The Quantity field indicates the number of units of this material required per unit or lot to manufacture the item.

Field/check box/radio button	Description
Lot Quantity Unit	These Lot Quantity and Unit radio buttons specify whether the quantity needed of the material is per lot or per unit. Lot is a set quantity for the whole job. If Lot Quantity is selected, the system considers the value entered into the Quantity field as the actual quantity needed. If Unit is selected, the system multiplies the Quantity field by the released quantity of the operation to derive the actual quantity needed.
Scrap Factor	The scrap factor is the percentage of expected scrap incurred during manufacturing for the component material. This percentage should be expressed as a decimal, e.g., 10% is entered as "0.1," not "10." When using a scrap factor the total quantity is calculated as: Total Quantity = Quantity * (1 - Scrap Factor)
Revision	The revision number of the item is specified in this field. This number helps you track which revision (authorized through Engineering Change Notices) the transaction affects. If you don't select a revision it defaults to the current revision.
Alt Group	This field is used when defining a group of alternate materials for an operation.
Alt Group Rank	This field identifies the priority rank for this alternate material within the alternate group of which it is a member.
Effective Date	If there is a date in this field, the planner ignores this operation or material previous to the effective date. The default of a blank date turns this feature off. This field appears only on the Current Operations and Current Materials screens.
Obsolete Date	If there is a date in this field, the planner ignores this operation or material after the date set. The default of a blank date turns this feature off. This field appears only on the Current Operations and Current Materials screens.
Backflush	This check box is selected if the material is to be backflushed. The check box status defaults to what was selected on the Controls tab of the item record.

Field/check box/radio button	Description
Backflush Location	This field can be updated only if the Backflush check box has been selected. If left blank, the system checks the hierarchy during the backflushing process to obtain a backflush location. At any point in the validation process that the system encounters a valid backflush location, it uses that location and ignore any other values set at higher levels. The order in which the system validates a backflush location is: • Current/Job/Production Schedule Materials • Floor stock locations for item at the operations work center • Items - Controls • Inventory Parameters

Current Materials – Costs menu option

The Costs menu option contains both the direct (unit costs) and indirect (overhead rates) costs of the material.

For your reference

For more details on costing, see the SyteLine: Costing Foundation Training Workbook.

Exercise 3.6: Add materials

In this exercise, you will create a new current materials record for DCBT-100.

Before you begin:

- Ensure you have completed Exercise 3.5 because it provides data or configurations for this exercise.
- Ensure you are logged in to SyteLine.

Exercise steps

1. Click the **Open** icon on the toolbar. The **Select Form** dialog box opens.
2. Type *current operations* in the **Filter** field.
3. Click the **web.CurrentOperations** list item.
4. Click the **OK** button. The **Current Operations** form opens.
5. Click the **Filter in Place** icon.
6. Type *[your initials]-DCBT-100* in the **Item** field.
7. Click the **Filter in Place** icon. The **[your initials]-DCBT-100** record for Operation 10 displays.

8. Click to select the **Operation 30** row in the slider grid.
9. Click the **Materials** menu option. The **Current Materials (Linked)** form opens.
10. Type *[your initials]-DCBTW-100* in the **Material** field.
11. Type *2* in the **Quantity** field.
12. Click the **Save** icon.
13. Click the **Window** drop-down arrow on the menu bar.
14. Click the **Close All** list item.

Customer orders

A company creates and uses a customer order (often abbreviated CO) when it sells goods or services. Each time a customer requests goods or services, a new customer order must be created.

The customer order record in SyteLine consists of two major parts:

- The header
- The associated lines/releases

In the planning process, the most important thing the order entry team needs to do is make good promises. As discussed previously, you need a good business process to ensure this happens. Part of that includes providing reliable information to order entry on item availability and production capacity.

After order entry has made the promise to the customer, you'll continue to monitor whether you're on track to keep that promise. This includes replanning the whole site, usually every night. At that time APS replans all customer orders, using the due date and the priority field to determine which orders to plan first.

APS plans each customer order line separately. You can get a projected date for the whole order, but that merely represents the projected date of the order line with the longest lead time.

Scenario: Customer orders

For our scenario, we will manually enter a customer order for the deluxe child bicycle trailer product. This order will provide the basis for our APS plan and schedule.

Exercise 3.7: Add a customer order

In this exercise, you will create a customer order for DCBT-100.

Before you begin:

- Ensure you have completed Exercise 3.6 because it provides data or configurations for this exercise.
- Ensure you are logged in to SyteLine.

Exercise steps

1. Click the **Open** icon on the toolbar. The **Select Form** dialog box opens.
2. Type *customer orders* in the **Filter** field.
3. Click the **web.CustomerOrders** list item.
4. Click the **OK** button. The **Customer Orders** form opens.

5. Click the **New** icon.
6. Type *[your initials]?* in the **Order** field.

   Note: Using your initials will make it easy for you to identify your orders later. By putting the question mark after your initials, the system automatically assigns the next available number. Make sure there is no space between your initials and the question mark.
7. Type *2* in the **Customer** field.
8. Click the **Save** icon.
9. Click the **Lines** menu option. The **Customer Order Lines (Linked)** form opens.
10. Type *[your initials]-DCBT-100* in the **Item** field.
11. Type *20* in the **Qty Ordered** field.
12. Verify the **Due Date** field displays the date **[4 days from today]**.
13. Click the **Save** icon.
14. Click the **Window** drop-down arrow on the menu bar.
15. Click the **Close All** list item.

Check your understanding

What is the purpose of modeling your environment for planning and scheduling?

(a) Maximize resource efficiency
(b) Ensure you have enough materials on hand to complete manufacturing processes
(c) Provide information about your resources, items, BOMs and settings to generate accurate plans
(d) Prepare to run What-If simulations to find the best planning option

What should you model as a resource?

(a) Anything that you need to schedule
(b) All people and machines used on the shop floor
(c) Materials, people, and machines required on an operation

Which of the following factors do you need to consider when modeling resources and resource groups for APS? Select all that apply.

(a) Identify resources to be scheduled
(b) Establish resource availability
(c) Set work center planning parameters
(d) Determine how resources will be allocated

When planning a manufactured item, what record does APS use to determine the time required to make the item?

(a) Item
(b) Operations
(c) Resources
(d) Resource Groups

Refer to Appendix B: Check your understanding answers on page 110 for answers to the check your understanding questions.

Lesson 4: Run APS and view the plan

Estimated time

0.5 hours

Learning objectives

After completing this lesson, you will be able to explain how to plan the site. In this lesson, you will:

- Describe the launch control process.
- Explain how to generate the Material Planner Workbench.
- Explain how to plan the site.
- Identify APS outputs used for analysis.
- Describe the basics of troubleshooting the plan.

Topics

- Running APS
- Planned orders
- Generating a workbench
- APS analysis
- Troubleshooting overview

Running APS

When you initially set up APS, and then on a scheduled basis (usually nightly), you use the APS Planning form to run the planning activity. This synchronizes all demands with the latest inventory, supply, shifts, and resources information.

When you run APS Planning, the system deletes all existing plan information and regenerates it using the latest data. In most situations, you will run APS Planning for a long horizon, such as one or two months. The APS Planning activity creates PLNs and exception messages.

To complete the planning activity, follow these steps:

- Submit the APS Planning activity to the background task queue.
- Verify the task completed with the Background Task History form.
- Review the APS output to view any warnings, errors, or blocked orders.

Run the APS Planning activity only at times when no other users will be saving changes to records that are included in the plan. Otherwise you may create data errors.

See Appendix D: Running APS on page 142 for additional information.

Planned orders

APS creates planned supply to meet demand. These suggestions are called "planned orders." They show up in the application as PLNs. APS calculates a JIT release date for each PLN.

On a daily or weekly basis, you will need to identify and release the PLNs when APS suggests you release them. Those you release will automatically be converted into actual orders.

If you are using MPS or PO requisitions, you will also want to identify and release MPS receipts and approved PO requisition lines.

Launch control

Our recommended practice for planning is releasing work to the shop floor in a JIT manner. We call this launch control, and it includes:

- Waiting to release planned orders until you've reached the JIT release dates APS has orchestrated, and
- Waiting to release planned work orders until you also have a full set of materials ready in inventory. Exceptions to this general rule are limited and clearly defined.

APS provides you the tools you need to see which planned purchase, transfer, and work orders to release and when to release them.

If you do not follow this recommended practice, and instead release work before it is needed, you will:

- Assign your resources to lower priority work, preventing your most important jobs from finishing on time
- Waste time ordering, processing, and storing more material than you need to
- Allocate materials to lower priority work, causing delays on more important jobs.
- Multiply the number of issues you have to address, making it harder to see and address the real problems in a timely manner

All of this means unnecessary delays and excessive lead times.

If you simply release when APS indicates it's time to release, you can avoid all of this. Customers who successfully implement APS can significantly cut their lead times. Launch control is one of the four key practices that leads to this result.

Generating a workbench

When you generate a workbench, you are actually selecting planned orders to put on your workbench to review and release to vendors and the shop floor. If you're using multi-site, you may also be releasing transfer orders to other sites.

The quick three-step process

You will identify and release your PLNs in a three-step process:

Step	Description
Step 1: Generate a workbench	The workbench displays all the PLNs, MPS receipts, and approved PO requisition lines that meet your criteria on your Material Planner Workbench using the Material Planner Workbench Generation activity.
Step 2: Review the workbench	Once a workbench has been generated, you can review and edit the orders using the Material Planner Workbench to: • Change order quantities • Combine orders • Change the order's due date • Select vendors for POs • Specify whether to firm planned POs, transfer orders, or production schedules into a new order or a new line on an existing order
Step 3: Release the PLNs	When you generate orders on the workbench, you are converting PLNs to actual orders with these statuses: • Job orders: Firm status • Production schedules: Planned or released status depending on parameter setting • Purchase orders: Planned status • Purchase requisitions: Requested status • Transfer orders

You will learn how to perform all three steps in this lesson.

Remember: The workbench needs to be used in conjunction with a launch control policy that helps you follow the four APS principles reviewed earlier.

Using the Material Planner Workbench Generation activity

The following is a summary of the fields used on the Material Planner Workbench Generation form.

Field	Description
Delete Planner's Batch	The workbench generation activity creates a separate workbench for each user ID/login. Further, when one user copies PLNs for an item, it prevents other users from copying any other PLNs for that item. This ensures that double copies are not created. When you delete a planner's batch, you clear the workbench for the specific user you are logged in as. You do not delete the workbench for anyone else. The choices for the Delete Planner's Batch field are as follows: • Yes: Deletes everything on the Material Planner Workbench and replaces it with the current generation. • No: Leaves all PLNs on the Material Planner Workbench. The generation will not create any PLNs for items already on the workbench.
Use Planned Data	The choices for the Use Planned Data field are as follows: • Yes: Copy APS PLNs, MPS receipts, and approved PO requisition lines onto the workbench. • No: Allows user to choose either the safety stock or cross-reference method. **Note:** When you “firm” an approved PO Requisition, the system will convert that requisition into a PO.
Source	The Source field identifies the source of the types of items to be included in the workbench generation.
	Records with release dates that fall after the date displayed in the Ending Date field are not included in the generation. Ending Date For example, if you usually release orders on a weekly basis every Monday, you will set this date to the coming Sunday. All orders that should be released before next Monday will be included in the generation set. Release dates are determined as follows: • Item type: Manufactured • Due date: Need date defined on order or calculated by planning engine • Release date: Calculated by the planning engine • Item type: Purchased or transferred Due date: Need date defined on order or calculated by planning engine Release date: Due Date - (Fixed Lead Time + Dock to Stock Lead time + Paperwork Lead Time + Variable Lead Time + Transit Time)
Planner Code	The Planner Code field is used to filter for items assigned to a specific planner.
Buyer	The Buyer field is used to filter for items assigned to a specific buyer
Ending DateOther fields	The Replenishment radio button and fields in the Detail section are used with other material planning methods.

The Order Action Report can also be used to see which PLNs need to be released. However, Infor recommends that you use the Material Planners Workbench instead of the Order Action Report.

The Material Planner Workbench Generation activity will also put MPS receipts and approved PO requisition lines on your workbench.

A workbench is assigned to a specific user login, and so each user needs to generate his or her own workbench.

An order can be placed on only one workbench at a time. You define which orders to put on a workbench using:

- Item source
- Release date
- Item planner code
- Item buyer

Exercise 4.1: Plan the site and create material planner workbench

In this exercise, you will run APS and generate the material planner workbench.

Before you begin:

- Ensure you have completed all prior exercises because they provide data or configurations for this exercise.
- Ensure you are logged in to SyteLine.

Exercise steps

Part 1: Run APS

Each time the APS Planning activity runs, all existing plan information and workbench information is deleted and regenerated using the latest data. If there are multiple learners in your organization taking this course or other any other courses which run APS at the same time, completing this exercise will affect all learners. You may want to coordinate with other learners before proceeding.

1. Click the **Open** icon on the toolbar. The **Select Form** dialog box opens.
2. Type *aps planning* in the **Filter** field.
3. Click the **APS Planning** list item.
4. Click the **OK** button. The **APS Planning** form opens.
5. Click the **Perform Single Site Plan** radio button.

6. Click the **Plan** button. A dialog box opens with the message, “Task Submitted”.

   This message means the task has been submitted in the background. In the next part of the exercise you will confirm that the task completed.

7. Click the **OK** button.
8. Click the **X** on the **APS Planning** form.

Part 2: Verify successful completion

1. Click the **Open** icon on the toolbar. The **Select Form** dialog box opens.
2. Type *background task* in the **Filter** field.
3. Click the **Background Task History** list item.
4. Click the **OK** button. The **Background Task History (Filter in Place)** form opens.
5. Click the **Filter in Place** icon on the toolbar.
6. Click the **Planning** row in the **Task Name** column.
7. Verify the **Return Status** field displays **Task Succeeded**. This means the task has completed successfully.
   Note: You may need to click the **Refresh** icon several times.

   Do not proceed to the next step until the Task Succeeded value appears. The Material Planner Workbench data will not populate until the planning task finishes.

8. Click the **X** on the **Background Task History** form.

Part 3: Generate workbench

1. Click the **Open** icon on the toolbar. The **Select Form** dialog box opens.
2. Type *material planner* in the **Filter** field.
3. Click the **Material Planner Workbench Generation** list item.
4. Click the **OK** button. The **Material Planner Workbench Generation** form opens.
5. Type *[your initials]* in the **Planner Code** field.
6. Click the **calendar** icon in the **Ending Date** field. A calendar displays with the current month and year visible.
7. Click the **[last day of the next month]** option. The **Ending Date** field populates.

8. Click the **Buyer** field.
9. Press **Delete**.
10. Click the **Process** button. A dialog box opens with the message, “ XX MRP Receipt(s) were processed.”
11. Click the **OK** button.

Part 4: View the workbench

1. Click the **Planner Workbench** button. The **Material Planner Workbench** form opens.
2. Click the **View** drop-down arrow.
3. Click the **Job** list item.
4. Click the **All Orders** radio button. The **Material Planner Workbench** form refreshes with all orders displayed in the grid.

Any exception messages are displayed in the grid. Scroll right to view the Exception Messages column.

5. Click the **Window** drop-down arrow on the menu bar.
6. Click the **Close All** list item.

View planning output

When the planning and workbench generation processes are finished, you can view the new planned orders and exception messages using the Material Planner Workbench or the Planning Detail form. It’s a good practice to check the workbench for exception messages each time you generate it.

Use the Planning Detail form to view exception messages separately from running the workbench. The form displays individual transactions created after you run APS Planning, and indicates the date you need each requirement.

This form is most useful in the grid view. The display is grouped to show all planning records related to a given item. The first record displayed for the item indicates the item's balance on-hand. The following records for that item represent jobs, sub-jobs, orders, etc. for that item. Any shortage or lateness is identified in the Exception Message column.

Exercise 4.2: View planning detail

In this exercise, you will view the planning output on the Planning Detail form.

Before you begin:

- Ensure you have completed Exercise 4.1 because it provides data or configurations for this exercise.

- Ensure you are logged in to SyteLine.

Exercise steps

1. Click the **Open** icon on the toolbar. The **Select Form** dialog box opens.
2. Type *planning detail* in the **Filter** field.
3. Click the **web.PlanningDetail** list item.
4. Click the **OK** button. The **Planning Detail (Filter in Place)** form opens.
5. Type *[your initials]- DCBT-100* in the **Item** field.
6. Click the **Filter in Place** icon. The **Planning Detail (Filter in Place)** form refreshes with all planning records for your item.
7. Click the **Maximize** button on the grid to display grid view.
8. Scroll right to view additional columns. The **Reference** column identifies the type of planning record and the **Exception Messages** column lists any issues.
9. Click the **X** on the **Planning Detail** form.

APS analysis

APS provides a wide array of analytical tools to identify potential order shipment problems and their likely causes. You will probably not use all the available tools in your daily work. Instead, you'll use the information that's most relevant for your role and responsibilities.

In this course we will introduce some of the essential information you can use to analyze your plans and schedules. In the subsequent role-based APS courses we provide more detailed information about how to use this information for resolving problems.

Order analysis

Of course, the first priority for your analysis is identifying orders that are projecting late. The primary order analysis forms and reports are:

- Demand Summary APS form
- Demand Detail APS form
- Demand Detail Chart APS form
- Delay Detail APS form
- APS Push Pass Setting form
- Exception Messages report

Following is quick reference for which tools can be used to analyze different types of order situations.

Order situation	Form/report
Customer order projecting late	• Demand Summary APS
Job order projecting late	• Demand Summary APS • Job Orders
Purchase order projecting late	• Purchase Order Lines
Identify causes of late orders	• Demand Summary APS • Demand Detail Chart APS • Delay Detail APS • APS Push Pass Setting • Exception Messages

Item analysis

If you determine that the cause of an order projecting late is that one or more items is missing or late, use the item analysis forms to get more information. The primary forms include:

- Component Shortages APS form
- Inventory Summary form
- Item Bottlenecks APS form
- Item Bottlenecks Report APS
- Planning Detail form
- Planning Summary form

Other forms and reports that may be useful for item analysis include:

- Component Shortages Report APS
- Exception Messages Report
- Item Bottlenecks Report APS
- Material Planners Workbench form/Ready flag
- Planning Detail APS form

Following is quick reference for which tools can be used to analyze different types of item shortage situations.

Item shortage situation	Form/report
Items causing delays	• Inventory Summary • Item Bottlenecks APS
Missing components for planned/released jobs	• Component Shortages APS • Material Planners Workbench/Ready flag
Lead time needs to be flexed	Expedite and move in exception messages: • Material Planner Workbench • Planning Detail Supply tolerance exception message: • Planning Detail
Time-phased view	• Planning Summary • Planning Detail

Resource/capacity analysis

Use the resource analysis forms to identify resources scheduled over capacity. Some of the analysis provides two types of each of the forms — one at the resource level and one at the resource group level. When planning in infinite mode, only the first resource in a group will ever be loaded, so it's usually more helpful to view the displays at the group level. The most commonly used forms include:

- Resource or Resource Group Utilization APS form
- Resource or Resource Group Load Profile APS form
- Resource or Resource Group Plan form
- Resource Gantt APS form
- Resource Group Bottlenecks APS form

Following is quick reference for which tools can be used to analyze different types of capacity situations.

Capacity situation	Form/report
Resource scheduled over capacity	• Resource or Resource Group Utilization APS • Resource or Resource Group Load Profile APS • Resource or Resource Group Plan • Resource Gantt APS

Capacity situation	Form/report
Capacity needs to be flexed	• Resource Group Utilization APS

Troubleshooting overview

This section introduces some general recommendations and common solutions for addressing shortages and late order situations. Specific troubleshooting steps and actions that you may take depend upon your role and responsibilities. In the subsequent role-based APS courses we provide details and examples of how to resolve issues.

Signs of trouble

The goal of APS is to help you ship on time. Trouble, therefore, is anything that would prevent you from shipping on time. APS provides you the following alerts to help you spot those issues.

- Red flags
  - Orders projecting late
  - Resources scheduled over capacity
  - Material shortages
- Exception messages

To ensure you stay on track, you need to review and respond to these alerts daily. Use the analysis forms described above to help you understand the sources of trouble.

Spotting trouble in APS versus MRP

If you are coming from an MRP background, you are probably accustomed to using past-due release dates as your signals that you have on-time shipment issues. The farther in the past you were supposed to release a supply, the higher priority it was.

However, APS does not work like that. APS will not plan into the past.

When APS sees that you don’t have enough lead time to get a component, it will push out the projected date for the supply. Because APS plans differently than MRP, this means that you will be looking at a different set of red flags that include:

- Orders with an APS estimated finish date later than the due date
- Resources that are overloaded
- Material orders that APS assumes will and can be expedited
- Materials that are causing delays

Once you understand those red flags, you will need to adjust your planning process to identify who is responsible to review each of the new red flags and act on the issues they uncover.

Reviewing exception messages

The planning activity generates exception messages. You can view these messages on the:

- Planning Detail form
- Exceptions Report

The key exception messages for use with APS are usually:

- Expedited
- Move In Receipt
- Move In Receipt (Tol)

- Move Out Receipt
- Rcpt Projected Late
- Receipt Not Needed

Infor recommends that planners and buyers filter the Planning Detail form each day for these exception messages, and take appropriate action to resolve each exception.

Guidelines for troubleshooting the plan

Each organization will develop its own process for troubleshooting. Here are some guidelines to help you get started.

Seek to improve the process

When you start to use APS, you may find that you keep encountering some of the same problems over and over. This usually indicates an opportunity to improve your business process. So instead of focusing all of your time on the urgent issues, budget time for the important task of seeking to improve your business process in order to eliminate or minimize the issue in the future.

This focus on improving processes applies to all the functional areas involved with APS activities. Make sure all areas are working together towards the same goal of generating a harmonized plan which allows you to make good promises and keep these promises.

View all of the key alerts every day

Your goal is to ship on time. Therefore, you want to know when you're off track and when APS is expecting you to take special actions to meet your shipping dates.

Which alerts you use depend upon your role and responsibilities. You do need to view all of those alerts relevant to your area each day. If you fail to look at all of them, you will eventually be blindsided. We recommend you schedule your time each day so you can view each alert or delegate the responsibility.

Be patient

When you first begin to plan with APS, you are likely to see a lot of problems. But if you faithfully review the trouble alerts each day and seek to improve the process, and if everyone is following all of the planning principles, then in a very short time, you will dramatically reduce the number of issues you see each day. Your patience and diligence will pay off.

Triage the issues

APS may present you with more problems that you can resolve in a day. This is especially the case when you first start planning with APS. Therefore, you will need some way to triage the issues you face so you can both make progress on the most important issues and view all of the key APS alerts each day. How you decide to triage is up to you, but we list our general suggestions below.

Determine your priorities, but avoid making your priority hierarchy too complex. Just give yourself a rough guide to start with. You will fine tune it as you go along.

Late order priorities

We recommend you divide your late customer orders into four periods:

- Critically past due
- Past due
- Due in the near future
- Due in the far future

Your priority will probably follow that sequence. However, we recommend you not focus on the urgent (orders past due) to the exclusion of the important (preventing future late order situations). As quickly as possible, you want to get to the point where you're preventing problems in the future instead of reacting to issues happening today. That means you need to spend time each day working on orders projected late in the future and on recurring problems that indicate a business process issue.

Capacity issue priorities

There are two types of capacity issues:

- Periods when you are scheduled between 80-100% utilization and therefore have a higher risk of not finishing on time
- Periods when you're scheduled over 100% utilization and APS is expecting you flex your capacity

Furthermore, not all resource groups are created equal — some resources delay far more demands than others. Resolving capacity issues at these resources will have a bigger impact on lead times in the shop as a whole.

We recommend you tackle the issues by splitting your time between what's urgent (near term overcapacity situations at bottlenecks) and what's important (preventing future overcapacity situations). Be sure to spend time each day working on capacity issues projected in the future and on recurring problems that indicate a business process issue.

Component shortage priorities

If you're following the four APS principles, you won't be releasing jobs to the shop floor that aren't ready. This means any component shortages for released jobs that you do encounter should be infrequent. And they will be caused by inaccurate inventory records, waste, or some inadvertent mistake.

As with the other issues, we recommend you tackle these issues by splitting your time between what's urgent (released jobs with shortages, especially those that are projecting late) and what's important (other jobs that are not late and planned jobs that are supposed to be released today or tomorrow). Again, allocate time each day working on planned jobs that are supposed to be released in the near future and on recurring problems that indicate a business process issue.

Common solutions

After using the APS tools to identify which jobs are projected late or past due, and what your resource issues are, it is time to determine the best way to resolve those issues. APS will not resolve the issues for you. Instead, you will need to determine the best solution for each situation.

After you decide how to resolve the issue, you can let APS know about the resolution so it can plan accordingly and so you don't see the issue in the future.

The actions listed below are common solutions to material and resource shortages, but there may be other actions you can take to resolve your issues.

Issue	Possible solutions
Purchased material shortage	• Expedite purchase orders • Build items in house • Find new sources of supply • Maintain a safety stock • Create a material reserve for unforeseen last-minute orders • Override an order's default priority so it gets the first shot at available materials • Change business process
Manufacturing resource shortage	• Work overtime • Move existing resources to work temporarily at bottleneck areas • Hire temporary resources • Outsource production • Plan to build appropriate reserves during slow times • Create a capacity reserve for unforeseen last-minute orders • Override an order's default priority so it gets the first shot at available resources • Change business process

Check your understanding

?

The recommended launch control process includes which of the following release policies? Select all that apply.

!

(a) Planned orders are released according to the JIT dates calculated by APS.
(b) Job orders are released to the shop floor in the order they are generated.
(c) PLNs are released as soon as they are created.
(d) Work orders are released only when all needed materials are ready and available in inventory.

?

The steps for planning the site and generating the workbench appear below in the wrong order. Reorder the steps from 1-8 to reflect the correct sequence.

!

Open the Background Task History form	
Open the Material Planner Workbench Generation form	
Click the Plan button	
Specify the Planner code	
Open the APS Planning form	
Verify Return Status field displays Task Succeeded	
Click the Process button	
Specify the ending date	

?

Match each of the following forms with the type of APS analysis it supports.

!

Form	APS analysis type
Resource Group Utilization APS	Item analysis
Demand Summary APS	Capacity analysis
Item Bottlenecks APS	Order analysis

What are the recommended guidelines for troubleshooting the plan? Select all that apply.

(a) Resolve all issues each day
(b) Try to improve your business processes
(c) Treat all late orders equally
(d) Be patient
(e) View key alerts every day
(f) Release all job orders as soon as possible
(g) Prioritize the daily problems

Refer to Appendix B: Check your understanding answers on page 110 for answers to the check your understanding questions.

Lesson 5: Next steps

Estimated time

0.5 hours

Learning objectives

After completing this lesson, you will be able to define the next steps for implementing APS. In this lesson, you will:

- Identify implementation preparation steps.
- Describe critical success factors.
- List additional APS learning opportunities.

Topics

- Preparing for implementation
- Critical success factors
- Additional learning opportunities

Preparing for implementation

As you prepare to implement APS, it's critical to ensure that your processes, teams, and data are harmonized and working together.

Process preparation

Each of the six key areas involved with APS in your organization must review its processes and procedures to prepare for using APS. Depending on your role and responsibilities, work with others in your area and in related areas to think about ways you can improve and prepare your processes.

Here is a recap of the key process requirements in each area for a successful APS implementation:

- Order entry: Demand management is robust, including forecasting, making promises, order prioritization, and related issue resolution.
- Purchasing: Supply management is working well, including purchasing policies and procedures along with related issue resolution.
- Inventory management: Inventory data is accurate and up to date.
- Engineering: Product structure has good integrity--in other words, good BOMs and routings exist and are used.
- Planning: Production control is in place. This includes scheduling, prioritization, and related issue resolution.
- Shop floor: Shop floor reporting is complete and correct.

Team preparation

All teams in these key areas must also be ready for implementation. This requires assigning ownership and responsibility for each element.

To start, make sure you have identified all the individuals in your area who need to be involved in both the implementation and the ongoing operation of APS. In each team, there must be a specific person named to own the integrity for each type of data used in the plan.

There must also be a specific person named to own the integrity of key processes in each area. Across the organization, this includes:

- Developing forecasts
- Making good promises to customers
- Releasing orders appropriately
- Identifying and addressing issues
- Communicating to shop floor operators what they need to work on next

Finally, all team members who will be responsible for providing data to APS or using APS output should take the additional role-based APS training relevant to their roles. These additional learning opportunities are described later in this lesson.

Data integrity

Like all planning systems, APS only works well when there is data integrity throughout the organization. This means that you need to ensure that APS has accurate inputs, and that all involved teams are ready to review and act on the APS outputs.

Review the inputs and outputs diagram in Lesson 1, and make sure that the individuals tasked with data integrity and processes have reviewed and verified each of these elements before you begin your APS implementation.

And to ensure that you not only start with valid data, but also maintain good data as you use APS, you will need business processes in place to continually verify the data.

It's very important to build this data quality and integrity into your daily operations.

It's also critical to model your environment accurately, including both materials and capacity. At the same time, don’t let the model get too complicated. Remember the 80/20 rule, and keep your focus on the resources and items that are most important to your business.

Critical success factors

Successful implementations of APS have shown that the biggest success factor is not the technology—APS is relatively simple to install and configure. The biggest success factors are:

- Maintaining realistic expectations
- Aligning your process
- Having the correct organizational motivation

When an organization fails to do these three things, the risk of an implementation failure (and the waste of the time and money associated with that) significantly increases. On the other hand, companies who do these three things are more likely to realize the benefits of APS, such as improved on-time delivery rates, reduced inventory levels, more efficient use of manufacturing resources, and better cash flow.

Maintain realistic expectations

The temptation when implementing a new tool is to believe that it will fix all problems. Some common false expectations about APS and what is actually realistic are listed in the table below.

Unrealistic expectation	Realistic expectation
This software will change the world for us.	This is a powerful tool that we can use to drive improvements in our business processes and on-time deliveries.
This software will fix all our business processes.	This tool will help expose flaws and limitations in our business processes, which gives us an opportunity to improve.
This software will automatically schedule everything.	APS gives us high visibility into potential problems at order entry time, allowing us time to adjust and modify as needed.
When we turn APS on, everything will work perfectly.	Even full, robust pilots can't fully replicate everything we'll encounter when we go live. It's going to take some time after we begin planning with APS to get the elements of our business process working harmoniously.
We can configure APS to produce perfect plans.	Seeking to configure or modify the software to produce hypothetically perfect plans that address every conceivable situation is futile. However, we can set up a process that produces excellent plans, enabling us to ship on time using lean principles.

When using any software, it is essential to have the right expectations. APS will not magically fix all your planning and scheduling issues. It is just a tool, and as when using any tool, you must follow certain guidelines to use it successfully.

Align your processes

Companies that use APS successfully have found that the hardest part of the implementation is making sure their business processes support what they're trying to do. If you want to use APS successfully, you need to align your business process.

This includes:

- Following the key principles and practices of APS as described previously
- Ensuring ownership and responsibility for each practice
- Seeking progress rather than perfection

These process alignment steps are critical to APS success. Make sure to work closely with an experienced consultant to identify how to best accomplish them in your business.

Ensure you have sufficient organizational motivation

The last success factor is the organization's motivation. And this is dependent on two things:

Why is your organization moving to APS?

Is your organization trying to solve a problem? Are you actively looking for a solution to improve your planning, or are you satisfied with your current planning system?

Are you looking to solve a planning problem or plan better? Or are you satisfied with your current planning process?

Many APS users have realized significant benefits in reduced lead times, reduced inventory, and increased on-time shipments. However, if you are happy with the results of your current planning methods, then you might want to consider if the switch to APS will provide the right return on investment. Only implement this new tool if it will help you solve a problem.

Is your organization ready to commit the necessary time and resources for success?

If you are looking to improve your planning and scheduling, you need to know that learning to use APS requires commitment and energy to:

- Revise appropriate business processes
- Obtain accurate data for critical resources and materials
- Train users

This investment of time and money will extend beyond your go-live date. Therefore, before embarking on an implementation, you need to make sure that management is willing to allocate the resources and time necessary to make it successful. If it is not a priority, your implementation has a high risk of failure.

Companies that are most successful with APS have selected APS to solve a specific problem, and are willing to commit the resources they need on an ongoing basis to make it work.

Additional learning opportunities

This course has provided a general introduction to APS and how it works. The next step is to learn how to use it in your daily activities, which varies depending upon your role and responsibilities.

Select from one or more of these role-based APS courses available on Infor Campus to take your learning to the next level.

SyteLine: Using Advanced Planning and Scheduling (APS) in Customer Service/Order Entry

This course teaches you key concepts of how to use Advanced Planning & Scheduling (APS) to support customer demands. You will learn the critical APS forms, fields, procedures, and processes used to incorporate customer orders into the planning process. You will also learn how to troubleshoot and adjust demand plans to ensure that that you can fulfill your customer promises.

SyteLine: Using Advanced Planning and Scheduling (APS) in Procurement

This course teaches you key concepts of how to use Advanced Planning & Scheduling (APS) to support the procurement process. You will learn the critical APS forms, fields, procedures, and processes used to plan and schedule your procurement activities. You will also learn how to troubleshoot and adjust plans and schedules to ensure that materials are available when needed to meet customer demands.

SyteLine: Using Advanced Planning and Scheduling (APS) in Production Planning

This course teaches you key concepts of how to use Advanced Planning & Scheduling (APS) to support the production planning process. You will learn the critical APS forms, fields, procedures, and processes used to plan and schedule your production activities. You will also learn how to troubleshoot and adjust plans and schedules to ensure that you can fulfill your customer promises.

Check your understanding

How should you prepare for implementing APS? Select all that apply.

(a) Identify individuals responsible for key processes
(b) Review and revise all BOMs
(c) Review processes and procedures in key areas
(d) Train all individuals throughout the organization in using APS
(e) Model every component of your operation
(f) Identify individuals responsible for data integrity

Match each of the following organizational actions with the critical success factor it supports.

Action	Critical success factor
Management commits needed time and resources	Realistic expectations
Key principles and practices of APS are followed	Organizational motivation
APS viewed as a powerful tool, rather than a solution to all issues	Process alignment

Refer to Appendix B: Check your understanding answers on page 110 for answers to the check your understanding questions.

Course summary

Estimated time

0.5 hours

Learning objectives

Now that you have completed this course, you should be able to:

- Explain the basics of how APS works.
- Identify important APS parameters and setup requirements.
- Explain how to model the site to use APS
- Explain how to plan the site.
- Define next steps for implementing APS.

Topics

- Course review

Appendices

The following are included in this section:

- Appendix A: User accounts
- Appendix B: Check your understanding (CYU) answers
- Appendix C: Setting up APS
- Appendix D: Running the APS planning activity

Appendix A: User accounts

Your instructor will assign you a student user ID from the table listed below to use for class exercises.

SyteLine: Fundamentals of Advanced Planning and Scheduling (APS) 01_0012105_IEN3734_SYT

Training Environment Entry Point (VM)	ID	User	Password
CSI CE Desktop	All	Infor[tenant type]	Infor123!

Application	ID	User	Password
Instructor Login (for course demos): Infor Ming.le	Instructor 01	in01@infor-edu.com	[varies]
	Instructor 02	in02@infor-edu.com	[varies]
Student Logins (for course exercises): Infor Ming.le	Student 01	st01@infor-edu.com	[varies]
	Student 02	st02@infor-edu.com	[varies]
	Student 03	st03@infor-edu.com	[varies]
	Student 04	st04@infor-edu.com	[varies]
	Student 05	st05@infor-edu.com	[varies]
	Student 06	st06@infor-edu.com	[varies]
	Student 07	st07@infor-edu.com	[varies]
	Student 08	st08@infor-edu.com	[varies]
	Student 09	st09@infor-edu.com	[varies]
	Student 10	st10@infor-edu.com	[varies]
	Student 11	st11@infor-edu.com	[varies]
	Student 12	st12@infor-edu.com	[varies]
	Student 13	st13@infor-edu.com	[varies]
	Student 14	st14@infor-edu.com	[varies]
	Student 15	st15@infor-edu.com	[varies]

Application	ID	User	Password
	Student 16	st16@infor-edu.com	[varies]
	Student 17	st17@infor-edu.com	[varies]
	Student 18	st18@infor-edu.com	[varies]
	Student 19	st19@infor-edu.com	[varies]
	Student 20	st20@infor-edu.com	[varies]
	Student 21	st21@infor-edu.com	[varies]
	Student 22	st22@infor-edu.com	[varies]
	Student 23	st23@infor-edu.com	[varies]
	Student 24	st24@infor-edu.com	[varies]

Appendix B: Check your understanding answers

This section provides the answers to the questions found at the end of each lesson.

Lesson 1: How APS works

Match each of the following essential APS practices with the key principle the practice supports. The key principles are: Keep data accurate, follow the plan, get ahead, and keep it simple.

Essential APS Practice	Key APS Principle
Make realistic promises when taking customer orders	Get ahead
Keep routings and BOMs up to date	Keep data accurate
Limit the number of constrained resources	Keep it simple
Release job orders only when you have all needed materials in inventory	Follow the plan
Use the 80/20 rule to decide what to model	Keep it simple
Record transactions timely and accurately	Keep data accurate
Review bottlenecks, capacity issues, and projected late orders daily	Get ahead
Release orders on time and according to APS release dates	Follow the plan

When does APS perform forward planning?

(a) On every demand
(b) When the demand due date is within a specified number of days from today's date
(c) When the demand due date cannot be met because the supply dates are later than the due date ✓
(d) After every pull pass

To meet the due date on a customer order, APS calculates you will need a part in three days, but there's no inventory available, and it will take 10 days to get the part in. APS will plan the customer order's projected date:

(a) Before the order due date
(b) After the order due date ✓
(c) Equal to the order due date

Which of the functional areas involved with APS are responsible for both providing valid data and for responding to APS outputs? Select all that apply.

(a) Planning ✓
(b) Purchasing ✓
(c) Inventory
(d) Order entry ✓
(e) Shop floor ✓
(f) Engineering

Lesson 2: APS parameters and setup

?

Which form is used to specify APS features and options?

!

(a) APS Options form
(b) Scheduling Parameters form
(c) APS Sites and Management form
(d) Planning Parameters form ✓

?

Which of the following statements about lead times are true? Select all that apply.

!

(a) Lead times for purchased parts are normally entered on the Items form. ✓
(b) Lead times for manufactured parts are normally entered on the Items form.
(c) Lead time values on the BOM are used to calculate precise projected dates for all items.
(d) Expedited lead times are specified only on an item-by-item basis.
(e) Expedited lead times can be applied on a global basis or an individual item basis. ✓

?

When might APS load a resource at over 100% of capacity?

!

(a) Never
(b) When the first pull pass of the planning process fails
(c) When the system is set for APS Mode
(d) When the system is set for Infinite APS ✓

Lesson 3: Modeling the site for planning

?

What is the purpose of modeling your environment for planning and scheduling?

!

(a) Maximize resource efficiency
(b) Ensure you have enough materials on hand to complete manufacturing processes
(c) Provide information about your resources, items, BOMs and settings to generate accurate plans ✓
(d) Prepare to run What-If simulations to find the best planning option

?

What should you model as a resource?

!

(a) Anything that you need to schedule ✓
(b) All people and machines used on the shop floor ✓
(c) Materials, people, and machines required on an operation ✓

?

Which of the following factors do you need to consider when modeling resources and resource groups for APS? Select all that apply.

!

(a) Identify resources to be scheduled ✓
(b) Establish resource availability ✓
(c) Set work center planning parameters
(d) Determine how resources will be allocated

?

When planning a manufactured item, what record does APS use to determine the time required to make the item?

!

(a) Item
(b) Operations ✓
(c) Resources
(d) Resource Groups

Lesson 4: Run APS and view the plan

?

The recommended launch control process includes which of the following release policies? Select all that apply.

!

(a) Planned orders are released according to the JIT dates calculated by APS. ✓
(b) Job orders are released to the shop floor in the order they are generated.
(c) PLNs are released as soon as they are created.
(d) Work orders are released only when all needed materials are ready and available in inventory. ✓

?

The steps for planning the site and generating the workbench appear below in the wrong order. Reorder the steps from 1-8 to reflect the correct sequence.

!

Open the Background Task History form	3
Open the Material Planner Workbench Generation form	5
Click the Plan button	2
Specify the Planner code	6
Open the APS Planning form	1
Verify Return Status field displays Task Succeeded	4
Click the Process button	8
Specify the ending date	7

?

Match each of the following forms with the type of APS analysis it support. The possible types are: Order analysis, item analysis, and capacity analysis.

!

Form	APS analysis type
Resource Group Utilization APS	Capacity analysis
Demand Summary APS	Order analysis
Item Bottlenecks APS	Item analysis

What are the recommended guidelines for troubleshooting the plan? Select all that apply.

(a) Resolve all issues each day
(b) Try to improve your business processes ✓
(c) Treat all late orders equally
(d) Be patient ✓
(e) View key alerts every day ✓
(f) Release all job orders as soon as possible
(g) Prioritize the daily problems ✓

Lesson 5: Next steps

?

How should you prepare for implementing APS? Select all that apply.

!

(a) Identify individuals responsible for key processes ✓
(b) Review and revise all BOMs
(c) Review processes and procedures in key areas ✓
(d) Train all individuals throughout the organization in using APS
(e) Model every component of your operation
(f) Identify individuals responsible for data integrity ✓

?

Match each of the following organizational actions with the critical success factor it supports. The possible success factors are: Realistic expectations, process alignment, and organizational motivation.

!

Action	Critical success factor
Management commits needed time and resources	Organizational motivation
Key principles and practices of APS are followed	Process alignment
APS is viewed as a powerful tool rather than a solution to all issues	Realistic expectations

Appendix C: Setting up APS

This appendix includes some additional information you may need to set up APS.

Planning Parameters form: IA recommended settings

The recommended settings for each field on the Planning Parameters form are displayed below. Infor suggests that you adjust the default setting to match these recommended settings as the best starting point.

Be sure to discuss your organization's specific needs with an experienced consultant using any different settings for these fields.

Parameter	Tab	Description	Setting
Use CO or Forecast	General	CO: The system plans customer orders but does not plan forecasts.	Both
Forecast Look Ahead	General	The parameter lets you set the number of days into the future to look for forecasts to be consumed by customer orders during a run of MRP Planning or APS Planning. When a customer order is entered in order entry, the system begins at the customer order item due date and looks this number of manufacturing days into the future for forecast orders to be consumed by the customer order.	0
Forecast Look Behind	General	The parameter lets you set the number of days into the past to look for forecasts to be consumed by customer orders during a run of MRP Planning or APS Planning. When a customer order is entered in order entry, the system looks this number of manufacturing days into the past (earlier than the customer line item due date) for forecast orders to be consumed by the customer order.	30
PO In	General	Number of days earlier than the current due date that the purchase order needs to be rescheduled before the system generates the Move Out Receipt exception message.	Null
PO Out	General	Number of days later than the current due date the purchase order needs to be rescheduled before the system generates the Move In Receipt exception message.	Null

Parameter	Tab	Description	Setting
Job In	General	Enter the number of days earlier than the current due date that the job needs to be rescheduled before the system generates the Cross Reference Special Receipt: Reschedule 99/99/99 exception message.	Null
Job Out	General	Enter the number of days later than the current due date that the job needs to be rescheduled before the system generates the Cross Reference Special Receipt: Reschedule 99/99/99 exception message.	Null
# Buckets for Report	General	The parameter maintains the number of buckets to display when directing output to a printer or file when printing the Planning Summary or Master Planning Reports.	12
30 Day Bucket Fence	General	Enter the number of days into the future that the system begins consolidating requirements into planned orders in 30-day buckets.	120
90 Day Bucket Fence	General	Enter the number of days into the future that the system begins consolidating requirements into planned orders in 90-day buckets.	360
MPS Plan Fence	General	Enter the number of calendar days into the future (starting from today) that the MPS Processor can automatically create and maintain MPS receipts based on the outstanding demands for an item. This value is used when the MPS Processor is run.	Null

Parameter	Tab	Description	Setting
Minimum Hours in Work Day	General	This number is used in calculating the internal manufacturing day (MDAY) calendar. The MDAY calendar is a list of valid manufacturing days between the MDAY Start and MDAY End date, using the default DSC scheduling shift and holiday definitions to generate the list. A manufacturing day is defined as those days set up in the MDAY Start and End range, greater than or equal to the hours in this field. For example, if Minimum Hours in Work Day is set to eight hours, and the total hours in the DSC scheduling shift for Monday is 7.5, Monday is not considered a valid manufacturing day.	8
MDAY Start		Enter the start day of the MDAY calendar. When generating the MDAY calendar, the system creates a calendar day for all the valid days (that are specified on the default scheduling shift) between the MDAY Start and MDAY End dates.	Enter start date
MDAY End	General	Enter the end day of the MDAY calendar. When generating the MDAY calendar, the system creates a calendar day for all the valid days (that are specified on the default scheduling shift) between the MDAY Start and MDAY End dates.	Enter end date
Check PO Requisitions	General	Considers an open PO requisition as supply	No (checkbox not selected)
Create PO Reqs from PLN	General	When generating orders in the Material Planners Workbench, purchase order requisitions will be created instead of purchase orders.	No (checkbox not selected)
Plan Planned Production Schedules	General	Select this field to include production schedules with Planned status when you run the MRP Planning or APS Planning activity.	No (checkbox not selected)
Use Scheduled Times In Planning	General	This field allows you to control whether MRP or APS generates the plan based on the jobs' current start and end dates or the Projected date calculated by the last run of the Scheduling activity.	No (checkbox not selected)

Parameter	Tab	Description	Setting
Apply Shrink Factor	General	Ability to increase the quantity of the planned job or order to compensate for expected loss prior to receiving an item to the stockroom	Yes (check-box selected)
Apply Scrap Factor	General	Select the field to apply the scrap factor as set up in the Job Materials form. If the scrap factor is not to be applied, clear the field.	Yes (check-box selected)
Plan Stopped Job Orders	General	Select this field to include job orders with a status of Stopped in the plan.	No (check-box not se-lected)
Plan Stopped Customer Orders	General	MRP will create a demand for customer orders in stopped status	No (check-box not se-lected)
Plan Planned Customer Order Lines	General	MRP will create a demand for customer order lines in planned status	Yes (check-box selected)
Plan Credit Held Cus-tomer Orders	General	MRP will create demand for customer orders on credit hold.	No (check-box not se-lected)
Move/Finish/Buffer Shift ID	APS	Enter the shift schedule to use when planning the Move, Finish, and Buffer times. If left blank, the APS Planner pro-cessing will be calculated using 24x7 availability. If a shift schedule is speci-fied, the APS Planner (and Scheduler) will use the shift definition to determine the real-time needed to achieve the Move, Finish, and Buffer times.	Null
Use Planning Output for Scheduling	APS	Select this field to specify that the system should copy the output from the APS Planning activity to the database tables used by the Scheduler output forms and reports. The system will copy this plan-ning output to the scheduling tables each time you run APS Planning. Also, when this field is selected and you run APS Planning, the system copies the opera-tion start and end dates calculated by APS to the Start Date and End Date fields on the operations forms.	Yes (check-box selected)

Parameter	Tab	Description	Setting
Include Existing Supply Usage in Ready Calculation	APS	This field is deselected by default. When deselected, only demands satisfied entirely from inventory are considered Ready (in the calculation of the Ready field.)	No (checkbox not selected)
Use Supply Usage Tolerance	APS	Select the field to use the tolerance factor defined in the Supply Usage Tolerance field. You can define a default, global tolerance value in the Supply Usage Tolerance planning parameter or you can define an item-specific tolerance value in the same field on the Items form. If the item-specific value is 0, the system uses the global tolerance value. The system applies the tolerance factor only if there are insufficient planned supplies and inventory available to satisfy the demand. If this is the case, the system can allocate planned supplies that are available within the tolerance time window to satisfy any remaining demand quantity.	Yes (checkbox selected)
Supply Usage Tolerance	APS	This field allows you to configure the APS system to be able to use a supply (such as a purchase order line item) to satisfy a demand that is due earlier than the supply is due. This field appears as an item-specific value on the Items or Multi-Site Items form and as a global value on the Planning Parameters form. A value of 0 entered on the Items form bypasses the item-level tolerance; the system instead uses the global tolerance value.	-10
Use Start Date	APS	Select this field to enable the Start Date field, which allows you to define the beginning of the planning horizon. If this field is not selected, the current date/time is used.	No (checkbox not selected)
Start Date	APS		Null

Parameter	Tab	Description	Setting
Start Date Offset (days)	APS	Specify a start date offset to shift the planning horizon into the past. The planning engine still uses the start date, if one has been defined, for calculations requiring knowledge of the current date, but allows demands to be planned in the past according to the offset amount. A positive number will shift the date into the past. Negative numbers are not allowed.	No (checkbox not selected)
For Firm Sub-Jobs	APS	By default, CloudSuite Industrial initializes sub-job start and end dates using the Lead Time Processor. By leaving the sub-job dates blank, the dates for firmed, and released, sub-jobs can be initialized by APS planning using backward planning from the need date. The Job Orders form will only allow blank start and end dates for firm sub-jobs if this parameter is selected. In all instances where CloudSuite Industrial would automatically populate start and end dates using lead time, the fields for firm sub-jobs will be left blank when the For Firm Sub-Jobs parameter is selected.	No (checkbox not selected)
For Released Sub-Jobs	APS		No (checkbox not selected)
For Planning	APS	This parameter controls whether APS will use expedited lead time when you run the APS Planning activity and/or when you perform a Get ATP or Get CTP operation. Select For Planning to use expedited lead time during a run of the APS Planning activity.	Yes (checkbox selected)

Parameter	Tab	Description	Setting
For ATP/CTP	APS	This parameter controls whether APS will use expedited lead time when you run the APS Planning activity and/or when you perform a Get ATP or Get CTP operation. Select For ATP/CTP to use expedited lead time during Get ATP/CTP operations. This option is not available on the What If Planning Parameters form.	No (checkbox not selected)
Demand Time (Hr) Demand Time (Min)	APS	Enter the hour and minute on which all demand orders and job/production schedule supply orders are due on the date they are due. The system does not track the time of day on dates associated with orders, but the APS planning functions plan at a detailed level of time. Therefore, you must specify the time so the planning functions have that information.	1 0
Supply Time (Hr) Supply Time (Min)	APS	Enter the hour and the minute on which supply orders (such as job supplies, purchase requisitions, purchase orders, transfer order supplies, and subcomponent demands) are available for the APS planning functions to allocate to demands. The system does not track the time of day on dates associated with orders, but the APS functions plan at a detailed level of time. Therefore, you must specify the time so the planning functions have that information. For example, if you enter 01:00, the system "sees" all purchase orders at 01:00. Enter the time in 24-hour format.	0 59
Manufactured Items	APS	Use this parameter to consolidate demand orders generated by APS into a single planned order (PLN), based on the consolidation option you select.	Days Supply
Purchased/Transferred Items	APS	Use this parameter to consolidate demand orders generated by APS into a single planned order (PLN), based on the consolidation option you select.	Days Supply

Parameter	Tab	Description	Setting
Max What-If Databases	Advanced APS	Enter the number of temporary copies of the planner database that can exist in the server's memory at the same time for APS planning functions. When you use the Get ATP or Get CTP functions, the system creates a temporary copy of the planner database in memory, on which it runs the calculations to determine the projected date for the order or line item. If multiple workstations run the Get ATP/CTP function simultaneously, the available memory on the server may be exceeded. Use this field to limit the number of concurrent database copies to within your server's memory capacity.	5
Infinite Purchased Items After (hours)	Advanced APS	Enter the number of hours (in the Plan Horizon) after which the system sets all purchased items as non-constraining. When this number of hours passes, no operation is delayed due to lack of a purchased item. However, the system still tracks material inventories. If you set this parameter to zero, the system ignores any constraints created by purchased items (both on-hand quantity and lead time constraints) during the entire Plan Horizon. The default setting is 999999, which indicates that all purchased items are constraining within lead time for the entire Plan Horizon. If the Infinite field is set for a particular Item, that Item will be considered an unconstrained item for the entire Plan Horizon.	999,999
Resource Group Buffer Scale	Advanced APS	During planning, the system multiplies this value by the Buffer In and Buffer Out values defined on every resource group. Increasing or decreasing the buffer times through this global parameter will allow you to uniformly adjust the slack in the plan without manually adjusting the buffer times on individual operations (resource group buffer time is added to the operation's Finish and Move hours). A value of 1.0 indicates no buffer scale.	1

Parameter	Tab	Description	Setting
ATP/CTP Pull Tolerance (days)	Advanced APS	During the Get ATP or Get CTP process, after the system push plans a demand to be late, it performs a final series of pull-planning iterations within a time window (bound by the demand's due date and the Projected date) to optimize the plan. This process divides the time window in half with each iteration until a feasible plan is found that is within the ATP/CTP Pull Tolerance number of days of the original due date. In the ATP/CTP Pull Tolerance field, enter the number of days to use as the system's limit of how close to the due date to attempt to plan the demand when planning with the Get ATP or Get CTP functions.	7
Planning Pull Tolerance (days)	Advanced APS	During a run of the APS Planning activity, after push planning a demand, the system performs a final series of pull-planning iterations within a time window (bound by the demand's due date and the projected date) to optimize the plan. This process divides the time window in half with each iteration until a feasible plan is found that is within the Planning Pull Tolerance number of days of the original due date.	7
Push Iterations	Advanced APS	Enter the maximum number of resource combinations the system will consider when push planning a single operation. We suggest you leave this field set to 0, to allow the system to find the fastest combination of resources to complete the operation earliest. When this field is 0, the system checks ALL combinations of resources in every resource group the operation requires. However, if you are concerned about system performance, you can limit the system from checking all combinations by entering a maximum value.	0

Parameter	Tab	Description	Setting
Pull Iterations	Advanced APS	Enter the maximum number of resource combinations the system will check while trying to pull plan a single operation. This parameter works like the Push Iterations parameter, except during pull-planning passes. We recommend you leave this parameter set to 0 to allow the system to find the fastest resource combination.	0
New Order Start Delay	Advanced APS	Enter the number of hours to delay the start of "new" incoming customer order lines, blanket releases, or estimate lines. "New" lines are those for which you have run the Get ATP/CTP process for the first time (and you haven't saved the record) or have been saved but the order date is still the current date.	0
Fixed Lead Time Reduction (Hours)	Advanced APS	Enter the number of hours that an item's standard Fixed Lead Time will be reduced in the APS plan. This parameter applies to all items.	0
Variable Lead Time Reduction (Hour)	Advanced APS	Enter the number of hours that an item's Variable Lead Time will be reduced in the APS plan.	0
Planner Trace Level	Advanced APS	Specify the level of trace messages to be written while the APS Planner is executing. The four trace levels and the trace flags that are enabled at each level are: • Minimal: Startup + Warnings • Limited: Startup + Warnings + Order + BOM + Item + Supply Switching • Extensive: Startup + Warnings + Order + BOM + Item + Event + Supply Switching • Full: Startup + Warnings + Order + BOM + Item + Operation + Final Resource + Event + Supply Switching + Time Fence The trace can be viewed on the View APS Planning Log form.	Minimal

Parameter	Tab	Description	Setting
Global Planning Mode	Advanced APS	In single-site mode, APS Planning regenerates the plan only at the local site. In global mode, APS Planning regenerates all plans, transfer supply orders, and transfer demand orders at all sites defined on the APS Sites and Alternative Management form. The sequence in which each site is planned is determined by the site priority defined at the local site where you run APS Planning. In Global Planning mode, each site will be planned in priority order. Demands for transferred items are planned at the supplying site when encountered at the appropriate point in the planning process. In Multiple Site Planning mode, each site will be planned in priority order. Demands for transferred items are planned using leadtime, with planned transfer orders replicated to the supplying site during planner post-processing. After all sites have been planned, the replicated transfer orders are planned incrementally.	Multiple Site Planning
Safety Stock Sequence Rule	Advanced APS	Use this field to specify a secondary sorting option that will be applied during the planner run. This applies only to items with the same low level code.	Greatest Percentage below Safety Stock
Use Latest Pull For Alternative Items	Advanced APS	The system uses this field during the pull planning phase of the APS Planning activity. It affects how the system plans an operation's materials that are members of an alternate materials group. Alternate materials are materials that have the same Alt Group ID in the operation's Current Materials record.	No
Apply Order Maximum Through BOM	Advanced APS	Select this field to apply order maximums to sub-levels of the BOM. Clear this field to restrict order maximums to the end item of a demand.	No (checkbox not selected)

Parameter	Tab	Description	Setting
Job & PS Supply Switching	Advanced APS	Select this field to allow the system to reallocate on-hand inventory and job/production schedule supplies to short-term demands that, during the APS Planning run, may have been satisfied by later job orders and production schedule releases. The system runs this process after each APS Planning run, in a "push" planning mode, starting from the current date/time.	No (checkbox not selected)
Planned Mfg Supply Switching	Advanced APS	This parameter works only when you run the APS Planning activity. It applies only to planned orders (PLNs) for manufactured items. The Planned Mfg Supply Switching parameter appears on the Planning Parameters form and on the Items form. For the system to apply the switching, you must enable the parameter on the Planning Parameters form and on each Item record you want to affect. Clearing the parameter on the Planning Parameters form disables it for all items.	No (checkbox not selected)
Plan Safety Stock before Forecasts	Advanced APS	Select this field to cause the planner to process safety stock before planning forecast orders, which prevents forecast orders from consuming safety stock.	No (checkbox not selected)
Allow Forecast BOM Supply Items	Advanced APS	Indicate whether a component of a forecasted end item be available for ATP/CTP	No (checkbox not selected)

Site level planning settings

APS has a number of settings that are applied on a site level. Determine how you will use each setting.

Use the information below to help you decide how to APS should plan and schedule.

The planning period

Guidance on the forms and settings used for different planning period elements is detailed on the table below.

For this	Consider how you will use this setting	On this form
MDAY calendar APS can only run for periods that are within the MDAY calendar.	Minimum Hours in Work Day MDAY Start MDAY End	Planning Parameters
	Shift ID (DSC) Description Starting Day and Time Ending Day and Time	Scheduling Shifts
Start date APS normally starts its planning period at the system's current date and time. If you want to turn off this constraint, update the following fields.	Use Start Date and Start Date, or Start Date Offset (days)	Planning Parameters
Planning Horizon The period APS plans	Plan Horizon (days)	Planning Parameters

Supply and demand dates

Use this information to help you set up dates for planning.

For this	Consider how you will use this setting	On this form
Receipt date for work orders Do you want to use the work order's end date or projected date as the date the items will be finished?	Use Scheduled Times in Planning	Planning Parameters
	Planning Analysis For Non-Scheduled Jobs (Use Job End Date, Use Projected Date)	Planning Parameters

For this	Consider how you will use this setting	On this form
Firm job projected dates. On firm jobs do you want the APS calculated projected date to reflect both resource and material constraints?	Preserve Pre-Released Production Dates **Note:** Selecting this will hide the impact of material shortages.	Planning Parameters
Requirement dates for materials Do you want APS to plan components to be available at the start of the job or at the start of the operation that uses the material?	Plan Materials at Operation Start	Planning Parameters
Default demand and supply times APS plans down to the minute. What time do you want to attach to the due date on orders you enter manually? (Those firmed up from planned orders will have the APS calculated time.)	Demand Time Supply Time	Planning Parameters

Supply availability

Consider the following questions to evaluate your supply settings.

For this	Consider how you will use this setting	On this form
Ready flag When you consider components for a manufacturing order to be available	Include Existing Supply Usage in Ready Calculation	Planning Parameters
PO requisitions Do you want APS to see PO requisitions as supply?	Check PO Requisitions Create PO Reqs from PLN	Planning Parameters
Production schedules Do you want APS to see planned production schedules as supply?	Plan Planned Production Schedules	Planning Parameters
Negative on-hand Do you want APS to recognize negative on-hand inventory quantities?	Consider Negative On Hand	Planning Parameters
Buffer for new planned orders When APS plans a demand in real time, do you want to make sure it does not create new planned orders that need to be released immediately?	New Order Start Delay	Planning Parameters

For this	Consider how you will use this setting	On this form
Supply switching Do you want APS to reallocate on-hand Inventory and job/production schedule supplies to short-term demands that, during the APS Planning run, may have been satisfied by later job orders and production schedule releases?	Job and PS Supply Switching	Planning Parameters
	Planned Mfg Supply Switching	Planning Parameters
Infinite supply Do you want APS to assume that you can get all supplies without restriction at some time in the future?	Infinite Purchased Items After (hours)	Planning Parameters

Exception messages

This form and setting can help you manage which exception messages are displayed.

For this	Consider how you will use this setting	On this form
Which exception messages do you want to report on?	Priority Level	Exception Message Priorities

Source linked supply

Use this form to help determine the settings for linked supplies.

For this	Consider how you will use this setting	On this form
Exception Messages When do you need to know that you need to reschedule a source linked PO or job?	PO Reschedule Tolerance Factor - In/Out Job Reschedule Tolerance Factor - In/Out	Planning Parameters
Source link sub-job dates Do you want APS to calculate the start and end dates of source linked sub-jobs? You can have APS do this in real time with Get ATP/CTP as well as during the APS Planning.	Calculate Job End Dates When Blank (For Firm Sub-Jobs, For Released Sub-Jobs)	Planning Parameters

Planning method adjustments

Here are some adjustments you may want to use when fine-tuning your APS planning operations.

For this	Consider how you will use this setting	On this form
Planning quantities To what decimal point do you want APS to plan each unit of measure?	Planning Precision	Unit of Measure Codes
Three-pass planning When APS has to do a three-pass plan, when do you want it to stop trying to improve the projected date?	ATP/CTP Pull Tolerance Planning Pull Tolerance	Planning Parameters
Resource combinations Do you want to limit the number of resource combinations APS checks when scheduling an operation?	Pull Iterations Push Iterations	Planning Parameters

Order prioritization

Optionally, you can set up variable order priorities. Use this information to decide how you will manage this.

For this	Consider how you will use this setting	On this form
Do you want APS to plan some order types before other order types?	Priority Level **Note:** We recommend you set all orders to the same priority. If, after having run APS for a number of months, you feel you need to adjust them, you can do so at that time.	APS Order Priorities
Are you going to prioritize customers, customer orders, or customer order lines? If so, make sure you have clear rules for when you will add these priorities.	Priority Level	APS Order Priorities
	Priority	Customer Orders Customer Order Lines

Database setup and log files

Some organizations may need some specialized database settings as shown below.

For this	Consider how you will use this setting	On this form
Get ATP/CTP databases How many planning databases do you need for Get ATP/CTP?	Max What-If Databases	Planning Parameters
Log files What level of trace messages do you want to be written while the APS Planner is executing?	Planner Trace Level	Planning Parameters

Orders to exclude from the plan

If you want to exclude certain orders from planning, use the following forms and settings.

For this	Consider how you will use this setting	On this form
Excluding customer orders that are stopped, planned, or on credit hold	Plan Stopped Customer Orders Plan Planned Customer Order Lines Plan Credit Held Customer Orders	Planning Parameters
Excluding job orders that are stopped	Plan Stopped Job Orders	Planning Parameters

Item level planning settings

You can adjust how APS plans each item in a number of areas. You need to set each adjustment for each of the items you are planning.

Use the information below to help you decide how APS should plan and schedule items.

All Items

The settings shown below apply to all items.

For this	Consider how you will use this setting	On this form
Supply tolerances Set the global supply tolerance to one day. Identify the items for which you will allow APS to allocate on-order quantities that are due to arrive more than one day after the need date. For those items, identify the maximum number of days you can expedite an order.	Use Supply Usage Tolerance Supply Usage Tolerance	Planning Parameters
	Supply Usage Tolerance	Items
Planned order consolidation Do you want to consolidate planned orders for purchased items, manufactured items, or both? Do you want to consolidate using a one-day bucket or something larger?	Planned Orders Consolidation Days Supply Planning	Planning Parameters
If you are using Days Supply, identify the bucket for each item.	Days Supply	Items
Do you care to see Days Supply sized buckets far out into the future? If not, consider how you will use the 30-day and 90-day buckets.	30-day bucket fence and 90-day bucket fence	Planning Parameters
Planned order minimums and multiples Do you have order minimums or multiples on any items?	Order Minimum Order Multiple	Items
Planned order maximums For those running in APS mode, do you need APS to break up orders for any item to schedule the full amount across many resources instead of just one? If so, updated the Order Maximum field on the Items form.	Order Maximum	Items
	Apply Order Maximum Through BOM	Planning Parameters

For this	Consider how you will use this setting	On this form
Component scrap Do you need to adjust your BOM's for scrap on any component item? This will increase the planned order requirements for the item.	Scrap Factor	Current Materials
	Apply Scrap Factor	Planning Parameters
Item shrink Do you regularly need to order extra quantities because of some type shrink, e.g. damage in transit, drop-off, etc.?	Shrink	Items
	Yield	Current Materials
	Apply Shrink Factor	Planning Parameters
Safety stock quantities Do you need to keep safety stock on hand for any items?	Plan Safety Stock before Forecast	Item/Warehouse
	Preservation Level Multiplier. Set it to "1"	Planning Parameters
	Time Fence Rule. Use "Lead Time" or "Accumulated Lead Time"	Items
	Safety Stock Sequence Rule	Planning Parameters
Short-term supply reservations Do you want to reserve supplies available (on-hand and on-order) inside a time fence for demands inside the same time fence? If you plan on releasing jobs well before the date they need to get started, you will need to use this setting.	Safety Stock	Item/Warehouse
	Time Fence Rule Time Fence Value	Items

For this	Consider how you will use this setting	On this form
Reallocating APS planned order supply allocations This parameter slows the performance of the APS Planning activity; therefore, in most cases you should not select it. It is designed to address a specific problem: when the parent item in a bill of material is defined with an Order Maximum value, and its component items are defined with Order Minimum values, the system may create excess planned orders. You can enable Planned Mfg Supply Switching to allow the system to reallocate the excess supplies appropriately for these items.	Planned Mfg Supply Switching	Items
Do you want to divide planned orders between multiple sources (vendors for purchased items, sites for transferred item)?	Vendor by Percentage, Site by Percentage	Source Rules

Items configured with the features and options functionality

These settings apply only to items that use features and options functionality.

For this	Consider how you will use this setting	On this form
Items flagged to be configured	Configured	Items
Planning by probability	Probable	Current, Job, and Production Schedule Materials

Summary of fields affecting planned order sizes

The fields that affect planned order size are listed in the table below.

Fields that affect planned order size	On these forms
Shrink Factor Order Minimum Order Multiple Order Maximum Days Supply Configuration Flag	Items

Fields that affect planned order size	On these forms
Apply Shrink Factor Apply Scrap Factor 30 Day Bucket Fence 90 Day Bucket Fence Apply Order Maximum Through BOM Planned Orders Consolidation Days Supply Planning	Planning Parameters
Planning Precision	Unit of Measure Codes
Scrap Factor Probable	Current, Job, and Production Schedule Materials
Vendor by Percentage Site by Percentage	Source Rules

Locations for available inventory

These settings apply to inventory location information.

For this	Consider how you will use this setting	On this form
Non-Nettable Stock APS does not plan inventory in non-nettable locations.	Non-Nettable Stock	Locations

Multi-site planning

If you have a multi-site setup, you need to determine if you will run APS on each site independently and use replication to coordinate demands and supplies between sites, or if you are going to run APS from a central site.

Pros and cons of each method are listed in the table below.

Site planning method	Advantages	Disadvantages
Individual site	Don't have to coordinate the running of APS across multiple time zones	Plan with lead times, not actual availability and capacity Create planned supply orders at future date after transfer order demands are received and APS runs at the remote site
Global	Plan with actual, real-time availability and capacity Create planned supply orders immediately at all sites	May have to coordinate the running of APS across multiple time zones

Planning a single order with multi-site

Get ATP/CTP works the same in multi-site as it does in single-site. The outcome when an item is sourced from a remote site is described in the table below.

Remote site setup/situation	Outcome from clicking Get ATP/CTP button
Remote site is set up on Planning Parameters form	Returns a projected date based on a Get ATP/CTP at the remote site plus transit time. The remote site returns the projected date.
Remote site is not set up on Planning Parameters form or remote site cannot be reached	Returns a projected date equal to lead time plus transit time.

APS will create planned transfer orders for transferred items that are sourced from other sites. You will see and be able to firm these orders up on the Transfer Order tab on the Material Planners Workbench.

Questions and settings

With a consultant experienced in multi-site, determine how you will plan. When you have finished discussing how you will operate in multi-site, use the information below to help you decide how to configure your settings.

For this	Consider how you will use this setting	On this form
Multi-Site Planning Mode	Global Planning Mode	Planning Parameters
Planning Sites	In transit time and other required fields	Inter-Site Parameters

For this	Consider how you will use this setting	On this form
Planning activity	Perform Single Site Plan or Perform Global Plan	APS Planning
Order of Planning in Global Planning	Priority	APS Sites and Alternative Management

Appendix D: Running APS

There are a number of additional factors you may need to consider when running the APS Planning activity, described in this appendix.

APS Planning form

The radio buttons and field on the APS Planning form are described below.

Radio button/field	Description
Perform Single Site Plan	Select this radio button to generate an APS plan only for the site where you are running the APS Planning (the “local” site). If an order requires a component produced at a remote site, the remote site will not plan the transfer demand order for that component. However, a transfer supply order is planned at the site where you are running the APS Planning activity, using lead time and transit time.
Perform Global Plan	Select this radio button to run APS Planning for all sites in your environment. **Note:** this option is available only if you have remote sites defined on the Sites tab on the Planning Parameters form. When generating a global plan across all sites in your multi-site environment, APS contacts each site in sequence of site priority and regenerates the local plan. APS removes and recalculates all planning information, creating or updating transfer supply and demand orders at each site. The local order priorities defined at each site determine how demands are satisfied at that site. If you are running a global plan, the planning database at each site must be running. If APS cannot contact a site (for example, if the planning database is down or there is a network problem), it uses the remote item's lead time and transit time to plan the transfer supply orders for those items.
Alternative	Select the production alternative (0) or a previously defined What-If Analysis alternative from this field.

Viewing planning warnings, errors, and blocked orders

APS may successfully finish planning yet encounter issues. Use the APS Planning and Scheduling Messages form to see all warnings, errors, and blocked orders from the last planning run.

Processes you may need to run before APS Planning

There are additional tools that you may need to run in the sequence shown in the table below. Notice that the APS Planning activity is the last activity in the sequence. That is because all the other processes update various inputs to the APS plan.

Sequence	Activity/Utility	Result
1	Rebalance Item Qty On Hand utility Rebalance Item Qty On Hand Reserved utility	Updates on-hand quantities
2	Global Priority Setting activity	Prioritizes your job orders; priority numbers used by the Scheduler **Note:** Used only if you are using the Scheduler.
3	Scheduling activity	Calculates start and end dates and times on manufacturing order operations; schedule feeds into APS plan **Note:** Used only if you are using the Scheduler.
4	MPS Processor activity	Creates MPS orders for MPS items with demands that occur after their MPS horizon and generates exception messages for MPS items; feeds into APS plan **Note:** Used only if you are using MPS.
5	APS Planning activity	Coordinates demand and supply

APS Resync All

Another utility you may want to run on a weekly or monthly basis is APS Resync All.

Use this utility to remove all APS input data and refresh it from current production data.

Because APS input data is synchronized continuously as updates are made to production data, this utility is not normally needed; however, you could run this utility weekly or monthly to, for example, guard against any updates being missed due to a SQL issue.

This utility is data-intensive. Run it during off-hours when there is minimal use of the database.

Run this utility only when:

- All other users are logged out of the application, and
- Data collection and other automated processes that can alter production data are not running.